



APPENDIX 2 - ENGINEERING SPECIFICATIONS

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ES1 Road and Footpath Works

Objectives

- O1 Engineering Standards - To provide adequate engineering standards for public domain areas, public road reserves and private access roads.
- O2 Uniformity - To ensure that there is a benefit to the public resulting from developments and the result is that the public is catered for by uniform infrastructure. Such infrastructure includes the road carriageway, footway, footpath, pavement, kerb and gutter, street trees, utility services, ancillaries and the like.

Definitions

- D1 Components - The road reserve is defined by the following components:
- Footway - the section of land between the kerb face and the property boundary
 - Footpath - the section of pavement within the footway dedicated for pedestrian and/or bicycle access
 - Kerb and gutter - the separation or interface between the footway and vehicular (road) carriageway
 - Road carriageway - the section of land dedicated for vehicular traffic
 - Ancillary items - any stormwater drainage asset, road/street furniture, edging, lighting, poles, services, signage etc., that forms part of the road reserve
- D2 Significant Developments - Major commercial and industrial developments, developments which occupy more than three sites across a street frontage, three or more allotment sub-divisions would be considered as "significant developments".

Controls

- C1 Requirements - All development involving demolition and/or construction will be required to satisfy the requirements of this Engineering Specification.
- C2 Footway Area Minimum Standards - The footway area shall be reconstructed to satisfy the following minimum standards.

Development	Requirement
1. Demolition only	Repair/ reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report
2. Alterations and additions	Repair/ reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report
3. New dwelling with existing footpath along frontage	Repair/ reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report
4. New dwelling with no existing footpath	Repair/reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report. Construct new footpath
4a. Dual Occupancy (duplexes, town homes, etc.) with no existing footpath	Repair/ reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report. Construct new footpath
5. Residential Flat Building, commercial and mixed developments	Repair/ reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report. Construct new footpath
6. Development under SEPP (Housing) 2021	Repair/reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report. Construct new or reconstruct existing footpath from site to nearest transport facility to AS1428.1 compliance
7. Significant Development other than those listed herewith	Construct new footway, footpath, road carriageway and ancillaries
8. Subdivision of one into two allotments	Repair/ reinstate damaged areas not reported in the Damage Report Form or Dilapidation Report. Construct new footpath, ancillaries and access road/s
9. 'Greenfields' site, subdivision of more than two allotments	Construct new footway, footpath, road carriageway and ancillaries. Construct access road/s. For Rhodes Peninsular Developments refer to Appendix 5F for additional requirements

Footway

Re-grading the grassed verge within Footway

- FW1 Footway Re-grading - The grassed verge within the footway is to be re-graded across the entire property frontage, for development types 7, 8, and 9, and in the following circumstances regardless of the development type:
- Grassed verge has cross-fall of more than 5% (1 vertical: 20 horizontal).
 - Matching in with constructed and existing driveway/s.
 - Existing grassed verge is difficult (undulating) for pedestrians to negotiate.
 - The construction of a vehicular crossing into the site results in unreasonable undulations in the footway or steep level changes.
 - The construction of a footpath pavement and/or re-alignment of the road carriageway require the footway to be raised or lowered to match.
 - Access difficulties for both pedestrian and vehicular traffic would occur without adjustment of the verge.
 - Walls and fences within the property boundary must match in with the new levels.

Minimum Footway Design Requirements

- FW2 Constraints - When adjusting footway levels, factors such as the levels of adjoining properties, the degree of pedestrian usage, existing trees, rock outcrops and other physical features and the alignment of existing road, must be taken into consideration.

- FW3 Minimum Design Criteria - The minimum design criteria are set out as follows:

- The longitudinal gradient of new footways shall not exceed 10%.
- The longitudinal change in gradient along footways shall not be greater than 1 in 10 with a minimum ease of 4.0 metres.
- A concrete footpath shall be constructed along the entire section of new footways.
- Where it is necessary to raise or lower the section of footway at the front of properties, the adjoining sections shall be re-graded to provide a smooth transition to the new levels.
- The construction of steps is generally not permitted unless the footway transition will extend more than 5.0 metres beyond the property frontage along the footway in order to achieve smooth transition.
- Where a footpath is to be installed and its longitudinal gradient exceeds 10%, over a distance of not less than 5.0 metres, steps may be installed subject to Council's approval.
- Grassed footway cross-fall shall be graded at 10% adjacent to the property boundary, where there is no formal or concrete footpath. The grassed verge near the kerb side shall not have a cross fall exceeding 5%. The absolute minimum footway cross-fall shall be at 1.0% (for grassed surfaces only). A footpath shall be constructed at all times. The footway shall be graded such that there will be no localised depressions which may cause water to pond or to allow the concentration of stormwater or directing stormwater into private property/s or creates a step or height difference which makes it difficult for the 85th percentile vehicle (B85) to open its doors on the verge side.
- Where access to properties is affected by re-grading of the footway, all work necessary must be done to ensure that satisfactory pedestrian and vehicular access is provided. This may necessitate transitions within the development. Where works are required on adjoining property/s, such as driveway adjustments, written consent from the property owner/s must be obtained.

Footway Material

- FW4 Material - Material to be used for filling of the footway shall be a minimum 75mm thick DGB20 or clean fill consisting of granular material free of vegetation, stumps, roots, rubbish, construction waste and other deleterious material.
- FW5 Rock Excavation - Where excavation is in rock, the rock shall be removed to a depth of 300mm below the finished surface level and replaced with clean fill.
- FW6 Topsoil - Minimum 100mm thick layer of topsoil shall be placed over the footway. Topsoil shall not contain more than 40% clay content. Clods in the topsoil shall not be greater than 50mm in diameter.
- FW7 Ground Cover - The footway shall be protected with couch, kikuyu or buffalo or other types of ground cover to match the existing surface type which provides good protection from surface erosion.
- FW8 Ground Cover Maintenance - The ground cover shall be maintained by the Developer for a minimum two (2) months after placement or laying.
- FW9 Developer Responsibilities - The Developer shall be responsible, at his expense, for the replacement of dead ground cover during the first two (2) month period after placement.
- FW10 Footway Acceptability - The footway will not be considered satisfactory if:
- The finished level of the footway adjacent to the kerb is below the top of the kerb or edging.
 - There is a step-down or uneven transition from any foot paving or driveway crossing to the finished level of the footway.
 - There are localised depressions in the footway which would cause the ponding of water or the concentration of stormwater runoff.
 - There are areas of eroded or dead ground cover.

Footway within Private Property

- FW11 Requirements of Footways - Generally footways for private access roads are not required unless access into the property/s, for example, as part of land subdivision including community title subdivision, requires the construction of an access road where its length exceeds 30 metres and no other safe access is made available for pedestrian thoroughfare.
- FW12 Width of Footways - Where a footway is required, the width of the footway shall not be less than 1.2 metres with service corridor/s located within the footway area in accordance with the Streets Opening Co-ordination Council Guidelines.

Footpath

Where Footpaths Required

- FP1 New Concrete Footpath - Full concrete footpath construction is required across the entire property frontage for development types 4, 4a, 5, 6, 7, 8, and 9 and beyond the development frontage for development type 6.
- FP2 Footpath Reconstruction - Full sections of footpath are to be reconstructed for development Type 3 to Type 9. Types 1 and 2, replacing broken pavements to ensure safety for pedestrian thoroughfare.

Footpath Minimum Widths

- FP3 Minimum Widths - Minimum footpath pavement widths are given in the following table.

Development type	Minimum Width (metres)
1, 2	Width to match existing but not less than 1.2m unless constrained by site conditions such as footway width being less than 1.2m
3, 4, 4a	1.2
5, 6, 7, 8 and 9	2.0
Regardless of above development type, the footpath is a shared route	Width as per Aust Roads Guidelines

Footpath Design Requirements

- FP4 Construction Material - All footpath pavements are to be constructed in plain full concrete. Pavers, bitumen, patterned concrete, coloured concrete, any other form of cosmetic treatment of concrete pavement or a combination of pavement types is not acceptable unless it has been approved as part of the streetscape planning of the area or the pavement is a replacement of an existing slab of similar material, pattern or colour.
- FP5 Footpath Cross fall - Footpath pavement cross-fall shall be a minimum of 1.0% and a maximum of 2.5% (1 in 40 for disabled access) or absolute maximum 5.0% (where disabled access is not required).

- FP6 Grading of Footpaths - Footpath cross-falls shall be graded away from the property boundary, towards the kerb and gutter or to an approved drainage system. Cross-fall must not be graded into private property/s.
- FP7 Concrete Minimum Strength - The minimum concrete strength shall be 25 MPa at 28 days and 80mm slump in accordance with **AS3600 – 1994, Concrete Structures Code**.
- FP8 Concrete Thicknesses - Concrete shall be installed at the minimum uniform thicknesses as given in the table below (except where footpath is across driveways, in which case, it shall be the same thickness as the driveway slab). Concrete footpath shall be laid on a minimum 100mm thick road base (compacted to minimum 98% maximum dry density) or 50mm thick and well compacted sand (compacted to a density index of not less than 65%).

Footpath Width (metres)	Slab Thickness (mm)
1.2 to 1.5	100
1.5 to 2.5	130
> 2.5	150

- FP9 Sub-Grade - The sub-grade shall be compacted and checked for uniformity and all irregularities made good prior to the pouring of concrete.
- FP10 Finish - The concrete shall be coving trowel finished only. All edges shall be rounded with a 75mm edging tool, with a 5mm radius.
- FP11 Joint Material - Contraction/expansion or construction joints shall be formed from 10mm thick compressible bituminous cork filler board (mastic jointing material).

- FP12 Joint Installation - Contraction/expansion joints shall be installed at the full depth of the slab at each side of vehicular crossing slabs, against concrete structures and at intervals given in table below.

Width of Slab (metres)	Distance C/E Joints (metres)
1.5	4.5
2.0	6.0
3.0	9.0
3.5	10.5

- FP13 Weakened Plane Joints - Weakened plane joints shall be 3mm wide and formed at intervals as given in the table below except for integral kerbs where they are to match joint locations in the slab.

Width of Slab (metres)	Distance WP Joints (metres)
1.5	1.5
2.0	2.0
3.0	3.0
3.5	3.5

- FP14 Concrete Curing - Concrete shall be subjected to a curing process for a minimum of seven (7) days. In the event of any damage or an unsatisfactory finish of the slab, replacement shall be carried out at the expense of the Developer at no cost to Council.
- The footpath slab must have a minimum thickness of 100mm. For footpaths with a width greater than or equal to 1.5 metres but less than or equal to 2.5 metres, the minimum slab thickness is 130mm, and SL72 reinforcement is required. Footpath with a width greater than 2.5 metres will require a slab thickness of 150mm and SL82 reinforcement. Reinforcement is to be centrally placed within the slab.

Location of Footpaths within the Footway

- FP15 Footpath Location - New footpaths shall be abutted against the property boundary or in the same alignment or continuation of the existing footpath or as deemed appropriate by Council's Asset Engineer, where no existing footpath is present. For reconstruction of existing footpaths, the same location as the adjoining footpath.
- FP16 Extension of Footpath - If the installation of a footpath results in the grassed verge to be less than 600mm in width, then the footpath shall be extended for the full width of the footway, that is, the concrete footpath shall be constructed from the edge of the property boundary to the back of kerb.

Signage

- FP17 Signage - Where footpath is to be used for both pedestrians and cyclists, adequate signage shall be installed at the commencement of the footpath indicating that the footpath is for shared pedestrian and bicycle usage.
- FP18 Signage Location - Signage shall be provided both on stems and painted onto the pavement at the appropriate locations, in accordance with the ***Aust Roads Bicycle Design Guidelines***.
- FP19 Signage Clearance - Signage stems shall have at least 600mm clearance from the face of kerb.

Kerb and Gutter

Where Kerb and Gutter is Required

- KG1 Kerb and Gutter Requirement - Regardless of the development type, kerb and gutter shall be constructed in the following circumstances:
- The proposed works includes the reconstruction or construction of an existing road or new road for access.
 - Kerb and gutter along an existing paved carriageway servicing the development is in an unsatisfactory condition or at unacceptable or non-standard level. For example, kerb height is more than 150mm or less than 150mm.
 - Kerb and gutter along an existing paved carriageway servicing the development does not exist.
 - Kerb and gutter along the property frontage is in an unsatisfactory condition or at unacceptable or non-standard levels.
 - Kerb and gutter along the property frontage does not exist.
 - The construction of kerb and gutter provides protection to the property/s from the inflow of stormwater off a public road or reserve.
 - The construction of kerb and gutter would prevent the likely erosion of the road shoulder as a result of vehicular and/or pedestrian traffic outside the proposed development.
 - Kerb and gutter is to be reconstructed as part of the footway and/or carriageway reconstruction.

Minimum Design Standards

- KG2 Minimum Design Criteria - Kerb and gutter shall be constructed to the following criteria:
- The minimum longitudinal gradient along the gutter shall be at 1%, where practical.
 - Cross fall from the centreline of the road to the invert of the gutter shall be at 3%.
 - The road maximum cross fall from the centre line of the road to the shoulder shall not exceed 5%.
 - The proposed kerb and gutter shall match in with existing kerb and gutter.
 - A section of not less than 1.0 metre of the existing kerb and gutter need to be reconstructed to ensure that there is a smooth transition between new and old works.
 - Existing crossovers and drainage outlets affected by the proposed works shall be replaced and extended to the new kerb face.
 - Kerb and gutter refer to standard 150mm high concrete kerb with integral gutter.
 - Other types of edge treatment such as rolled kerb and gutter, mountable kerb and gutter, dish crossing, sandstone blocks or brick kerbs may be used if it has been approved as part of the streetscape planning of the area or the edging type is a replacement of an existing material, pattern or colour.
 - Kerb only is not to be used where the cross fall of the road diverts stormwater to the kerb.
 - Kerb and gutter shall have tool joint at 3.0m intervals and mastic joints at 12.0m intervals.

Road Carriageway

General

- R1 Reference - The road carriageway refers to the section of the road reserve dedicated for vehicular traffic flow. This includes any private access roads (e.g. rights-of-way).

When road carriageway is to be reconstructed

- R2 Carriageway Construction - The road carriageway is to be reconstructed in the following situations:
- Half road reconstruction for development type 7 and full road construction for development types 8 and 9.
 - Regardless of the development type, a Damage or Dilapidation Report confirms that the road has been significantly damaged as part of the development or is in very poor condition or failure at the time of construction/development, half road reconstruction will be required.
 - Regardless of the development type, the existing road carriageway outside the property frontage is at a cross fall which results in the scraping of vehicles when a vehicular crossing has been installed for the property. Full road slab construction will be required for local road concrete pavements to Council's standard concrete road pavement specification.

Road carriageway construction

- R3 Cross-fall - Construction of the road carriageway is to have the following cross-falls:
- Cross-fall to be graded from the centre or crown of the carriageway to the gutter.
 - One-way cross-falls may only be permitted for narrow roads and one-way trafficked carriageways of less than 5.0 metres in width.
 - Offset crowns are not acceptable.
 - Super-elevation is not normally provided but shall be a maximum of 6% at bends.
 - The minimum cross-fall for asphaltic concrete carriageway pavement shall be 2.5% to 3% (with absolute maximum of 5%).
- R4 Longitudinal Gradient - Longitudinal gradient of the road carriageway shall be as follows:
- For bitumen pavements, the minimum longitudinal gradient of the road carriageway shall be 1%.
 - For concrete pavements, an absolute minimum fall of 0.5% is allowed, construction accuracy permitting.
 - For all pavement types, the general maximum longitudinal gradient shall be 12.5% for residential (local) roads. For other road classification, in accordance with the ***Aust Roads Pavement Design Manual***.
 - For all pavement types, the general maximum longitudinal gradient shall be 8.5% for local distributor roads.
 - An absolute maximum longitudinal gradient of 16% may be permitted depending on adequate sight distance of vertical curves.
- R5 Matching in with Existing - The proposed road pavement shall match in with the existing pavement with regard to adequate vertical eases and horizontal transitions. This may require the reconstruction of a portion of the existing road pavement to match in smoothly with the new works.
- Where reconstruction of existing road way is proposed, a pavement design shall be undertaken to Council's satisfaction based on geotechnical investigation.
 - For all road rectification works, treatment types shall be submitted to Council for approval prior to implementation.

- R6 Where reconstruction of existing road way is proposed, a pavement design shall be undertaken to Council's satisfaction based on geotechnical investigation.
- R7 For all road rectification works, treatment types shall be submitted to Council for approval prior to implementation.
- R8 Pavement Design Standard – All Road Pavements shall be constructed as a rigid road pavement comprising:
- Road pavement constructed in concrete shall be minimum 50MPa compressive strength at 250mm uniform thickness with two layers of reinforcement fabric (SL82) top and bottom 50mm cover over a 150mm lean mix concrete of 5MPa compressive strength.
 - Verification of sub-grade suitability shall require submission of a geotechnical report with relevant core sampling tested and reported.

Ancillaries

Pedestrian Crossing Ramps

- A1 Location - Pedestrian crossing ramps shall be constructed at each kerb return opposite the extension of the main footpath and in the kerb opposite the extension of a public pathway.
- A2 Design and Installation - Pedestrian crossing ramps shall be designed and installed as per **AS1428.2**.
- A3 Raised Tactile Pavers - Raised tactile pavers shall be installed on pedestrian ramps located in high pedestrian environments such as town centres and positioned as per **AS1428.2**.
- A4 Tactile Paver Material and Colour - Raised tactile pavers shall be of grey coloured concrete or black coloured polyurethane type material or as specified as part of the streetscape planning for the area or the tactile paver is a replacement of an existing paving of similar material, pattern or colour.

Vehicle Crossings and Laybacks

- A5 Minimum Standards - The minimum standards for the construction of driveways and laybacks across the footway are provided in Section **ES2 Vehicular Access**.

Stormwater drainage

- A6 Installation of Piped Drainage - For development types 7, 8, and 9, piped stormwater drainage must be installed to cater for at least the minor system flows as described in Section **ES3 Stormwater Management**.
- A7 Minimum Standard - The minimum standards for stormwater drainage system design are given in Section **ES3 Stormwater Management**.

Traffic Calming Devices

- A8 Minimum Standards - Traffic calming devices such as thresholds, slow points, speed humps, chicanes, and the like are to be designed in accordance with ***Aust Roads Guide to Traffic Engineering Practice, Part 10, Local Area Traffic Management.***

Steps

- A9 When Steps Required - Steps should be avoided whenever possible. However where necessary, they shall be installed along all pedestrian-only footpaths where the longitudinal gradient of the pathway is sufficiently steep to warrant its installation or that the installation of a transition ramp cannot be extended adequately to achieve a smooth transition due to site constraints. "Steepness" is measured in accordance with the Building Code of Australia requirements or as determined by Council's Asset Engineer, where BCA requirements cannot be met.
- Steps are also to be installed where the longitudinal gradient of the footpath exceeds 1 in 4, over a distance of not less than 5.0 metres.
- A10 Material - Steps shall be constructed in reinforced concrete, pre-cast units or prefabricated metal and supported on-ground and are to be painted in highly visible paint to delineate change in height. Suspended types are not preferred.
- A11 Preference of Ramps over Steps - Ramps are favoured over steps whenever possible. To permit access for disabled persons, steps are considered as a last resort and only when ramps cannot be installed to comply with the disabled access code due to physical constraints.
- A12 Handrails - Where the level change between the steps and/or footpath and the adjoining surface level, at a distance of 1.5 metres or less, exceeds 500mm, handrails are to be installed. Handrails are to be designed in accordance with **AS1428.2**.

Utility services

- A13 Satisfactory Provision of Services - Satisfactory arrangements for sewer, water, gas, electricity and telecommunication services must be provided for all developments including "greenfield" sites and land subdivisions.
- A14 Approval - Approval will not be granted unless satisfactory service provisions can be provided.
- A Section 73 Certificate under the Sydney Water Act 1994 must be obtained for "greenfield" sites and new land subdivisions. This Certificate must be provided prior to the release of the Subdivision or Occupation Certificate.
- A15 Provision of Electricity Services - Direct access to electricity supply must be provided for all "greenfield" sites and new land subdivisions. New electrical sub-stations, generators, kiosks, and turrets servicing the development must be located wholly within the property boundary. New electrical sub-stations, generators, kiosks and turrets must not be located on public land.
- A16 Undergrounding of Electrical Services - Development types 7, 8 and 9 including "greenfield" sites and new land subdivisions must have electricity supply provided underground. Including public domain areas, existing overhead power lines are to be re-installed as underground cables.
- A17 Light pole and Power pole and vehicular crossing - A minimum of 300mm clearance between the face of the pole to the edge of vehicular crossing and the face of the kerb, unless specified otherwise by Council.
- A18 Gas Services - Reticulated gas supply, where available and allowable, shall be provided for all "greenfield" sites and new land subdivisions.
- A19 Certification - A Compliance Certificate from each utility service authority must be provided certifying that its requirements for the satisfactory provision of the service have been met before release of Subdivision or Occupation Certificate.

- A20 Dedicated Service Corridor - For "greenfield" sites and new land subdivisions, a dedicated service corridor, conduit and easement shall be provided to service each allotment created.
- A21 Relocation of Services - If, as part of the proposed development, mains, services and poles need to be relocated, it shall be carried out at the Applicant's expense to the satisfaction of Council and the relevant utility authority, as required.
- A22 Developers' Responsibility - It is the responsibility of the Developer to ensure that all care has been exercised to prevent damage to any public utility, e.g. gas, water, sewerage, electricity and telecommunication. The Applicant will be responsible for any damage caused by him or his agents, either directly or indirectly.
- A23 House Service Alterations - Alterations to house services shall be carried out only by qualified tradesperson.
- A24 Location of Utility Services -The locations and depths of utility services under a road reserve shall be installed in accordance with the latest revision of Guide to Codes and Practices for Streets Opening prepared by **New South Wales Streets Opening Coordination Council** resolution.
- A25 Liaising with Relevant Utility Service Authorities - In particular relation to "greenfield" sites, new land subdivisions, and where a new or existing public road is to be reconstructed or constructed, the Developer must liaise with the relevant authority, arrange and pay all costs and fees associated with providing street lighting, underground electrical power, telecommunication services, and adjustment of surface fittings, pits, etc. affected by the propose works.
- A26 Engineering Plans - All underground services and services in general must be detailed on engineering plans, submitted and approved by Council prior to development approval.
- A27 Public domain lighting – All public light shall be designed and constructed to the requirements and specification of the City of Canada Bay Council.

Street Trees and Landscaping

- A28 Installation of Street Trees - Street trees are to be installed across the entire property frontage, for development types 7, 8, and 9 and where the footway and road is to be reconstructed regardless of the development type.
- A29 Suitable Species - Council's Landscape Architect is to be consulted in regards to suitable plant species.
- A30 Location - Street trees are to be planted at an offset distance of not less than 600mm from the face of kerb to the centre of the tree. Trees are to be planted at the distances tabulated below.

Location	Distance (metres)
From any road intersection (tangent point/kerb return)	10
Between tree centres	6
From the top of driveway wings	2
From signs, posts	1
Street Lighting Poles	10

- A31 Edging - For concrete pavements, no timber edging is required around the landscape islands. For surface types other than concrete pavements (e.g. turf or bitumen), the perimeter edging around the landscape island, shall be timber 30mm wide by 100mm deep. Landscape islands shall be square in shape, 1200mm wide by 1200mm long.
- A32 Bedding Material - For concrete pavements, bedding material shall be "arboresin" or equivalent. For bitumen pavements or grassed areas, bedding material shall be mulch. Colour shall be as specified by Council's Landscape Architect at the lodgement of the Development Application (DA).
- A33 Submission of Detailed Landscape Plans - Detailed landscape plans are to be submitted and approved by Council prior to development consent.

ES2 Vehicular Access

Objectives

- V1 Uniformity - To ensure uniformity in the design and construction of vehicular crossings in the City of Canada Bay Local Government Area.
- V2 Safe and Convenient Access - To ensure that safe and convenient vehicular access can be provided to and from parking spaces for all properties.

Definitions

- V3 General - Vehicular access or driveway crossing refers to the formal access for vehicles (85th percentile vehicle in accordance with **AS/NZS2890.1:2004 Off Street Car Parking Code**).
- V4 Formal Access - Driveway refers to the formal platform between the kerb line and the property boundary by which vehicular access is provided.
- V5 Exclusions - In this Section, vehicular access does not include internal access roads, (both vehicular and pedestrian access) within the property boundary, suspended structures and the like, which would be assessed separately on its merits and will require separate Development Application (DA) approval.
- V6 Ancillary Works (Minor) - Ancillary works (Minor) refers to any works other than vehicular crossings and laybacks, such as footpath, kerb, ramp construction or reconstruction, stormwater drainage connection (Road Opening Permit for stormwater connection is required), and the like within the road reserve. These works could be undertaken in conjunction with or separately from vehicular crossing and footpath construction.
- V7 Ancillary Works (Major) – Ancillary works (Major) refers to any works other than Ancillary Works (Minor), such as stormwater connection to the street underground drainage system, construction and reconstruction of pipes, pits and kerb lintels, as well as pavement within the road reserve. These works necessitate a separate application and consent under Civil Works in the Public Domain Application.

Application

- V8 Applications to Council for the location and construction of a vehicular crossing or ancillary works must be submitted.
- No construction work will be permitted until formal written consent is obtained from Council following the submission of the Application. An application for Vehicular Crossing Construction and Ancillary Works must be submitted and consented to by Council for all proposed construction and reconstruction of vehicular access, and this applies to:
- All Developments Applications involving residential, commercial, and industrial development,
 - All Complying Development Certificate application, and
 - Where a stand alone formal crossing(s) or cross-over (layback) is proposed.
- V9 Consultation - Prior to the lodgement of a Vehicular Crossing Application, please ensure that:
- The Applicant has consulted with Council's Planning Section, in regard to heritage, trees, LEP and DCP matters and has been given advice that a driveway/vehicular crossing at the specified location is permissible; and that
 - The Applicant has read this Specification and is fully aware of any limitations and/or constraints, which may preclude the approval of a driveway.

Statutory and Design Requirements

Persons who can undertake Construction

- V10 Approved Persons - Vehicle crossings can be constructed by either Council's Construction Team or a Private Contractor, to ensure that works are built to a reasonable standard and uniformity.

Property owners may appoint their own Private Contractor to construct the driveway. However, to ensure that the quality of the work is maintained and public safety is not compromised, the nominated Contractor must have extensive experience in concrete works especially in the construction of vehicle crossings with current public liability insurance cover and must be approved by Council.

A Private Contractor, nominated by the Property Owner, may be approved by Council to carry out the works following submission of the following:

- A copy of the current public liability insurance is provided to Council. The insurance cover shall be for a sum not less than \$A20,000,000 for a period of six (6) months, and nominating on the policy the City of Canada Bay Council as indemnifying them against public risk claims, arising during the construction of the crossing or as a result of the construction of the crossing;
- A copy of the Contractor's current concrete license and record of previous work are provided to Council; and
- A Security Deposit is paid to Council by the Contractor, as set down in Council's Fees and Charges lodged prior to any works commencing. This Deposit is refundable upon satisfactory completion of the works, at the discretion of Council.
- An additional non-refundable administration fee, as set down in Council's Fees and Charges shall also be paid. This fee covers the processing of the driveway application and three (3) site inspections. Note this fee applies to each driveway applied for per property. If more than one driveway is to be constructed, the same fee must be paid for any additional driveway to be processed.

If all of the above requirements are not satisfied, the application may be rejected and a further fee (10% of the Security Deposit) will be required before reconsideration.

For ancillary works such as the construction of a concrete footpath, an administration fee is payable. A refundable deposit is also required.

Issuing Driveway Levels

- V11 Written Permission - No work shall commence without written permission from Council.

The Contractor is to submit a driveway design appropriate for the site, complying with **AS/NZS2890.1:2004**. This design must be approved by Council in writing before any construction can be carried out.

In exceptional circumstances, e.g. due to omission of information on the part of the applicant and notwithstanding the above advice,

Council may consider that the construction of a driveway at the proposed location or site is inappropriate or impractical. The Applicant will then be advised accordingly and all but \$A50 will be refunded.

Supervision and Inspections

- V12 Supervision - Works are to be carried out in accordance with this document, relevant Council Specifications and Australian standards.

No work is to be carried out except under the supervision and approval of Council or its nominated representative.

Council shall meet with the nominated Contractor and/or Owner to carry out mandatory inspections for the purpose of:

- Formwork inspection - at least forty-eight (48) hours notice must be given for checking timbering for alignment, reinforcement placement, level and stability and sub-grade preparation.
- Final inspection - the Owner or Contractor shall notify Council for a Final inspection once the concrete has been poured and finished, formwork removed and area restored. At least twenty-four (24) hours notice must be given.

To arrange for appropriate inspection times, bookings shall be made through Council's Customer Service Section.

If more inspections are required above the mandatory inspections due to any omission or unsatisfactory work, including the storage of materials, insufficient barricading and site safety, on the part of the owner or contractor, or remedial actions that Council deems necessary to achieve compliance, an additional fee shall be charged, as defined in Council's Fees and Charges Schedule.

Final Approval

- V13 Notification in Writing - Council will notify the owner in writing of the Final Approval of the work. Any defects found during the Final Inspection must be remedied. This is the responsibility of the owner and therefore it is advised that final payment not be made to the Contractor until such approval has been received.

Once the driveway is completed and Council is satisfied with the works, a refund of the crossing bond can be arranged. The Applicant shall apply in writing to Council.

Adjustments to Utility Services

- V14 Service Alterations - Any alteration, which may be necessary, for the construction of the vehicle crossing, to any water, sewer, gas, electricity, telecommunication, stormwater system, and other utility services is to be arranged by the contractor with the relevant utility authority.
- V15 Protective Box Adjustments - The protective boxes over any hydrant, gas cocks, stop valves, sewer lines, and the like shall be adjusted by the contractor, in consultation with the relevant utility authority, so that they are flush with the finished surface.
- V16 Physical Location - The contractor is responsible for the physical location of all utility services likely to affect the proposed works. The contractor is liable for any damage to service utilities. The contractor is required to contact "Dial before you dig" (ph 1100) before any work commencing.
- V17 Associated Costs - Any cost incurred as part of the adjustment will be borne by the owner.
- V18 Conflicts with Stormwater Conduits - Where an existing or proposed house stormwater pipe traverses the proposed driveway crossing, and there is insufficient cover over the pipe, the section of pipe shall be converted to a galvanised steel 125 x 75 x 5 or 200 x 100 x 6 rectangular hollow section (RHS) across the footpath, to achieve adequate cover. Alternatively, relocate the pipe, where possible, away from the driveway and in the direction which allows adequate drainage

Road Design Affecting Driveway Levels

- V19 Proposed Design - Where a proposed design has been prepared by Council to alter the carriageway and/or footpath fronting the property concerned, the driveway works shall be carried out in conformity with the proposed design. Information for this purpose shall be supplied by Council or its representative.
- V20 Proposal to Modify Street Levels - Where Council has a proposal to modify the area at the front of the property but a proposed design has not yet been prepared, the work shall be set out in accordance with the best available information and in accordance with this Document and any other relevant standards, on the understanding

that if Council should carry out future construction works to a proposed design, which may involve the alteration to part or whole of any of the constructed work, then a contribution may be payable had no work been carried out, will be levied and must be paid by the property owner, in accordance with the Roads Act, at the time when the Council proposed works are to be carried out.

Public Safety and Provision for Traffic

- V21 Pedestrian and Vehicle Safety - The contractor is responsible for the safe passage of pedestrian and vehicular traffic. During the progress of the works, all necessary warning notices, barricades and lights must be installed, in accordance with **AS1742.3-2009, Traffic control devices for works on roads**.

Where the works require traffic control, the contractor will be responsible for appropriate traffic control devices being put in place, including necessary lamp signage, maintenance and the like in accordance with **AS1742.3 - 2019**.

Where works are to be carried out on roads of a "Collector Road" status or higher, and obstruction to traffic is unavoidable, a Traffic Management Plan must be submitted with the application to Council, endorsed by the Police and relevant authorities e.g. Transport for NSW, for approval. This plan is to be prepared in compliance with **AS1742.3 - 2019**. A minimum notification period not less than ten (10) working days shall be provided prior to the commencement of works.

These ancillaries should be included in the total cost towards the works.

- V22 Liability - The contractor shall be liable for any accident, damage or injury to persons or property resulting from the work. In this regard, the Contractor must have appropriate and current public liability insurance to this effect.

Compliance with Other Regulations

- V23 Compliance - Works shall be carried out in compliance with The Clean Waters Act, The Roads Act, The Motor Traffic Act and the Occupational Health and Safety Act, and any other Acts as deemed relevant.

Hours of Construction/ Demolition

- V24 Hours of Construction/Demolition - The hours of construction/demolition shall be restricted to 7:00am to 5:00pm, Mondays to Fridays and 8:00am to 1:00pm Saturdays, with a total exclusion of work on Sundays and Public Holidays.
- V25 Non-offensive Works - Non-offensive works, where power operated plant is not used such as setting out, surveying, plumbing, electrical installation, or site cleaning by hand shovel and site landscaping, is permitted between the hours of 1:00pm to 4:00pm Saturdays, at Council's approval.

Important Note: The Protection of the Environment Operations Act 1997 may preclude the operation of some equipment on site during these permitted working hours.

Preparation and Excavation

- V26 Minimum Excavation - The Contractor shall excavate to the full depth required for the specified thickness of the proposed crossing slab.
- In rock, clay or unstable foundation material, additional excavation shall be carried out and a sub-base of sand, or other approved material, to a minimum depth of 75mm consolidated thickness, shall be provided.
- V27 Compaction - The sub-grade or sub-base shall be adequately compacted to the required minimum depth of 150mm to the following requirements:
- Sands - density index of 70% in accordance with **AS 1289.E3.1** where the compaction test is in accordance with **AS 1289.E5.1**.
 - Material other than sand - dry density rates in accordance with **AS 1289.E4.1** of more than 95% where the compaction test is in accordance with **AS 1289.E1.1** (standard).
- V28 Adjacent Surface - The surface adjacent to the proposed driveway shall be trimmed to conform generally to the levels and cross-fall similar to the adjoining area, free of trip hazard, unless otherwise instructed by Council's Engineer. This shall include lifting and resetting footpath lawns where necessary.

Driveway Dimensions and Alignment

- V29 Maximum Driveway Slab Widths - The width of any crossing to a property must be kept to a minimum. Excessively wide vehicle crossings will not be approved because it:
- compromises pedestrian safety by encouraging vehicles to cross the footway at greater speeds,
 - minimises the area for pedestrian refuge,
 - encourages illegal parking on the footway,
 - detracts from passive streetscape by increasing the amount of visible hard paving, and
 - reduces on-street parking spaces.

Refer to Part B3 of DCP for crossing and driveway widths.

- V30 Driveway Slabs for Commercial Premises - Commercial premises, properties requiring two-level entries, or where traffic is heavy (for example, the property frontage to a main road with heavy vehicular traffic movement); the above widths may be increased at Council's discretion. For example, service stations may be granted wider driveway widths based on swept vehicle paths and submission of a Traffic Report.
- V31 Wheel Strips - Wheel strips **are not acceptable** because they do not provide sufficient protection to the public footway. Constant wearing by vehicular traffic may cause rutting in the grass verge which compromises the safety of pedestrians using the footway.
- V32 Vehicle Crossings to be Perpendicular to Kerb Line - Vehicle crossings are to be constructed perpendicular to the kerb line. In special circumstances, to reduce disturbance to traffic flow along a major road, an angled driveway may be permitted subject to the approval of Council's Traffic Engineer.

Driveways off Narrow Roads

- V33 Splays and Widened Driveways - Splays are not to be used. The maximum crossing slab width shall be 3.0 metres. Council does not encourage the construction of wide crossovers where it is unnecessary as it increases the risk to pedestrians and diminishes the opportunity for on-street parking.

Vehicle crossings are generally required to be constructed perpendicular to the kerb line. This encourages vehicles to slow down whilst entering properties.

In special circumstances, the driveway may be widened to prevent vehicles from driving onto the nature- strip or where safer access and egress can be provided.

Where the width of the road carriageway is less than 5.5 metres, the driveway may be permitted to be widened, subject to a check using a standard swept turning path of the 85th Percentile vehicle (B85).

Permitted Number of Driveways per Property

- V34 A maximum of one (1) vehicle crossing for vehicle access is permitted per site. See Part B of DCP for further information.

Driveway Locations

- V35 Prohibition - Access shall be located so that sight distances are not affected by existing structures such as street trees, earth mounds, bus shelters, and other physical features. Council may not approve the location of such an access if the location will adversely affect sight lines or detracts from the streetscape of the area.

Vehicle access will not be permitted at the following locations and circumstances:

- At Council's discretion
- Where the proposed vehicle access is likely to diminish on-street parking
- To a major road if reasonable access can be gained from another public road of lower classification
- Opposite or within 6.0 metres of a median opening in a major road

- Within 25.0 metres of a signalised intersection*
- Within 9.0 metres at non-signalised intersections*
- At localised depressions** ("sag" points)
- At the entrance into a basement or sub-floor level regardless of whether it is in a localised depression or not***

The above distances are measured perpendicular from the face of the kerb of the intersecting street (or prolongation of the kerb line or tangent point if curved).

***Important Note 1:** Access may be permitted provided that it can be demonstrated, to Council's satisfaction, that it is safe to do so and that extenuating circumstances exist, for example, where this is the only location where vehicular access can be gained.

****Important Note 2:** At sag locations in the road, if the proposed driveway could result in the driveway becoming a spillway for stormwater entry or stormwater ponding here, unless an appropriate hydraulic model (using either Manning's Equation or HEC-RAS, whichever being appropriate and based on the 1%AEP design storm) is used to determine the water surface profile to demonstrate that the driveway profile has a design crest which prevents stormwater from entering into the premises and/or that the stormwater runoff can be safely conveyed along a suitably designed overland flow path, then the driveway will not be approved.

*****Important Note 3:** All driveways to basement or sub-floor areas, whether it be located in a sag or not, will not be approved unless a catchment analysis for the 1%AEP design storm is undertaken to determine the stormwater runoff and then an appropriate hydraulic model (using either Manning's Equation or HEC-RAS, whichever being appropriate) is used to determine the water surface profile, to demonstrate that the proposed driveway will offer adequate protection from stormwater entering into the premises. Refer Section **Overland Flow and Flood Studies** for further details.

V36 Minimum Length of Parking Bay - A driveway will not be permitted if the proposed parking space within the property between the boundary and building line is less than 5.5m in length, causing a parked vehicle to encroach onto the footway reserve.

V37 Sight Lines - Vehicle crossings shall be located so that minimum sight distances to traffic and pedestrians can be provided. Sight distance to pedestrians shall be met by providing clear sight lines in accordance with **AS/NZS 2890.1 - 2004, Parking Facilities, Part 1: Off-street car parking.**

Vehicle crossings should also not be located where existing trees or power poles may obstruct sight lines or where it is too close to the root zone of trees. Similarly, the driveways shall not be located too close to power poles where undermining may occur.

V38 Removal of Obstructions - In certain situations, Council may grant approval for the removal or relocation of these structures, where there is no alternative location.

In this circumstance, approval must be sought from Council's relevant officers who may include Asset Engineer, Traffic Engineer, Tree Preservation Officer and Planners. All costs associated with such work shall be borne by the Owner.

V39 Stormwater Pits - All driveways must be clear of existing stormwater inlet pits. The removal or reduction in the length of the pit lintel or grating is not acceptable, as this would reduce the rate of stormwater collection. However, excluding existing pits located in a depression (sag), if the hydraulic characteristics of the drainage system are not made less efficient, the relocation of the pit may be permitted. In this circumstance Council will undertake the works of pit relocation or nominate an approved Contractor. All costs associated with the relocation of the pit shall be borne by the Owner.

V40 Street Trees - Driveways must not be located over or near trees to be retained. The minimum clearance to trees will depend on factors such as the proximity to the trees root zone and sight lines.

Existing Crossings

- V41 Retention of Existing Crossings - Existing crossing slabs and laybacks may be re-utilised if:
- They are in the correct location, set at the correct levels and in reasonable condition, **and**
 - its retention is NOT contrary to this Document.
- V42 Removal of Existing Crossings - Existing crossings and layback must be removed, where the crossing slab and layback is made redundant. It shall be completely removed and the footway area and kerb/gutter restored to Council's satisfaction.
- V43 Owner's Cost for Removal of Redundant Crossings - Any existing un-used crossing(s) and/or layback(s) must be removed and the kerb/gutter and footpath reinstated/restored at the owners expense to Council's satisfaction.

Driveway Levels and Gradients

- V44 Design – The Applicant shall submit a detailed longitudinal driveway profile starting from the centerline of the roadway or the crown of the road including 2m from the lip of the gutter to the road pavement, which will show dimensions, levels and gradients, transition and headroom clearances for Council consent.

Note: Existing roadway levels (e.g., centreline or 2m from the lip line) shall not be assumed and shall be reflected on the survey plan prepared by a registered surveyor.

- V45 Absolute Maximum Longitudinal Gradient - The absolute maximum longitudinal gradient of driveways shall not exceed 25% within the property and 10% in the footway (unless justification can be provided to vary these grades). Where the gradient exceeds 18%, a check is required using standard vehicle template (B85) to ensure that it does not scrape.

If a pedestrian footpath exists or is required, the gradient across the footway must not exceed 2.5% (1 in 40) where the footpath meets the driveway.

The location of the footpath is to match existing or 450mm offset from the property boundary or as directed by Council's Asset/Traffic Engineer if no footpath currently exists. It may be necessary to transition existing footpaths both sides of the driveway to align with the new driveway such that cross fall across the footpath is maintained at not more than 2.5% (3% absolute maximum). Driveways are to be graded to fall towards the kerb wherever possible.

- V46 Changes in Gradients - Changes in gradients shall not exceed 12.5% algebraically (1 in 8) in a crest and 15% (1 in 6.7) in a depression (sag) to prevent vehicles scraping or bottoming.

Changes in grades in excess of 12.5% (crest) and 15% (sag) will require the introduction of transitions. A minimum transition of 2.0 metres in length will be required (a 1.5 metres transition may be permitted if a template is overlaid onto the driveway profile which demonstrates that it will not scrape).

- V47 Compliance – Driveways shall be designed to match in with the topography of the site, and shall comply with **AS/NZS 2890.1 - 2004, Parking Facilities, Part 1: Off-street car parking**.

Council, at its discretion, may not approve a Driveway Application on the basis that the information provided to Council, at the time of approval, was in error or there are deficiencies in the information provided for Council to make an informed determination at the time of the assessment.

Driveway Material

- V48 Concrete - All vehicle crossing slabs and laybacks are to be constructed in plain concrete with a minimum compressive strength of 32MPa at 28 days.
Ready mixed concrete conforming to **AS1379-1973** shall be used. The Contractor is to arrange for certificates by the manufacturer to be given for all concrete delivered and shall be able to produce these to Council's Representative upon request.
- V49 Cosmetic Pavement - Construction of driveways, within the road reserve, using brick pavers, coloured, stained, stamped or patterned concrete, pebblecrete, or any other cosmetic material other than plain concrete **will not be approved**. Only standard plain concrete finishes to vehicle crossings will be permitted.

- V50 Base - Clean sand shall be used, free of any deleterious material, compacted and screeded to a smooth finish. Bedding thickness is to be a minimum of 50mm uniform thickness.

Compact sand bedding to a minimum density index of 70 as per **AS1289 5.6.1**.

For road base (DGB20), the standard shall be **AS1289.5.2.1**.

- V51 Filling Material - Where filling under the proposed concrete is necessary, such filling shall consist of granular material of maximum size of 40mm and shall be spread in layers of a maximum thickness of 150mm and consolidated to provide a 95% compaction when tested under the modified proctor method.

Expansion and Construction Joints

- V52 Expansion and Construction Joints - Shall be provided to the full depth of the slab, at each side of the slab and where required or as directed by Council.

The joint shall be filled with a 12mm thick bitumen impregnated material such as a compressible mastic board. Other types of jointing material can only be used with the approval of Council.

Expansion joints shall separate the concrete apron from the driveway area that it adjoins, in line with the leading edge of paths.

- V53 Concrete Footpaths - Concrete footpaths shall have expansion and tooled (dummy) joints installed at the widths as indicated below:

Slab Width (metres)	Tooled Joint (metres)	Expansion Joint (metres)
1.2	1.2	3.6
2.0	2.0	6.0
3.0	3.0	9.0
3.5	3.5	10.5

Finish

- V54 Concrete Finish - The concrete surface shall be finished true and even, free from air and stone pockets, depressions and projections. The concrete shall be tamped and screeded to the correct surface levels and shall be given an even non-skid finish.

The crossings are to be cove finished in the direction of the kerb and gutter unless it has a gradient steeper than 1 (vertical) to 5 (horizontal), where it shall be finished with a wooden float and then grooved (ensure grooves do not create tripping hazard to pedestrians). The path section behind the apron shall be coving trowel finished across the path. All edges of the slab shall be rounded with a 50mm edging tool.

- V55 Slip Resistance - In general, driveways should have a slip resistance appropriate for the pavement slope in accordance with **AS3600**.

Minimum Driveway Slab Thickness

- V56 Council Standards - All work is to be carried out in accordance with Council's standard details, Construction Specification for Concrete and Restoration Works, where applicable.
- V57 Minimum Thickness of Gutter Crossings - Gutters and gutter crossings (laybacks) shall have a minimum uniform thickness of 200mm or 220mm for commercial and industrial sites.

- V58 Minimum Thickness of Driveway Slabs - The minimum crossing slab thickness shall be in accordance with the following table:

Development Type	Minimum Slab Thickness (millimetres), Reinforcement
Residential	125, SL72
Commercial	200, SL72
Industrial	200, SL72
Residential Flat Buildings	150, SL72

- V59 Minimum Thickness of Footpaths - Footpaths shall be a minimum of 100mm in thickness with reinforcement to match existing pathways unless under driveway in which the thickness shall be the same as for the driveway or as directed otherwise by Council

Non-Compliance

- V60 Non-compliance - Failure to comply with the conditions of this Document will result in the approval to carry out construction of the crossing being revoked. If works are constructed in non-conformity, Council will issue an order to have the area restored or remedied.
- V61 Recovery of Costs - The cost for reinstatement of non-complying works shall be recoverable by Council, for example, deducted from the Security Deposit held by Council.

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ES3 Stormwater Management

Objectives

- SWM1** Uniform Guidelines - To provide uniform guidelines and applying control systems to achieve consistency, in the assessment and conditioning of Development Applications, in relation to stormwater runoff from all development types.
- SWM2** Minimise Adverse Impact to Properties - To minimise any adverse impact on properties caused by stormwater runoff from all developments types.
- SWM3** Minimise Impact on Water Quality - To ensure that the water quality of receiving waterways is not adversely affected by the discharge of pollutants such as nutrients and pathogens, from stormwater runoff as a result of development.
- SWM4** Uniform Controls - To ensure that uniform stormwater controls are applied throughout the whole of the City of Canada Bay Council Local Government Area.
- SWM5** To incorporate effective measures to improve the water quantity and quality of stormwater leaving the sites, where likely, potentially negative impacts have been identified.
- SWM6** To prevent adverse effects on the flood peak at any point upstream or downstream of development.
- SWM7** To emulate the runoff characteristics back to under the "State of Nature" condition.
- SWM8** To ensure that adequate stormwater management system to be provided within the development site.
- SWM9** Minimise alteration of flow regimes and velocities to avoid adverse impacts on the proposed development or other properties.

Controls

- SWM10** Controls - To achieve the above objectives, the following controls are applied:
- For controls where the site is within a Flood Planning Area, refer to the Flood Planning Section of Council's Development Control Plan.
 - The provision of safe overland flowpaths within developments and on public land.
 - The definition of floodways for major storms within developments and on public land.
 - The provision of controls such as on-site stormwater detention, community basins and the like and on-site retention systems to reduce and control stormwater runoff.
 - The application of alternative methods of merit based stormwater control and conveyance devices.
 - The removal of flood effected development from known floodways and the prohibition of future developments in such floodways.
 - The provision of minimum free-boards for assigning floor levels to reduce the risk of flood damage to property/s.
 - The installation of pipe/channel systems to minimise hazard to pedestrian and vehicular traffic caused by uncontrolled surface stormwater runoff.
 - The installation of water quality control devices such as trash screens, gross pollutant traps, water quality ponds and the like and encouraging the use of water sensitive urban design to protect the quality of receiving waters.
 - Depending on the development type and general site fall, the following stormwater control types would be applicable.

Development Type	Site slope/ Site drains to	Control Type
1	Front boundary	E, G
	Rear boundary	B, C, D, E, G
2	Front boundary	A*, F**
	Rear boundary	A+, B***, C, D, F**
3, 4, 4a, 4b	Front boundary	A, D++, F
	Rear boundary	A+, B***, C, D++, F
5, 6, 7, 8, 9	Front boundary	A, D++, F, H
	Rear boundary	A+, B***, C, D++, F, H
2, 3, 4, 4a, 4b, 5, 6, 7, 8, 9	Waterfront property	E, F, H

Key:*Development Type*

- 1 - Demolition Only
- 2 - Alterations and additions
- 3 - New dwelling with existing footpath along frontage
- 4 - New dwelling with no existing footpath
- 4a - Dual occupancy
- 4b - Townhouses, villas etc.
- 5 - Residential Flat Building (RFB), commercial and mixed developments
- 6 - Development under SEPP (Housing) 2021
- 7 - Substantial Development, Major alterations and additions
- 8 - Subdivision of one into two allotments
- 9 - "Greenfields" site, subdivision of more than two allotments

Stormwater Control System

- A - On-site Stormwater Detention System (OSD)
- B - On-site Retention System or absorption system (OSA)
- C - "Charged" system
- D - Mechanical Pump-out System
- E - Scour and erosion control system/devices
- F - Rainwater Re-use System
- G - Sediment Pond
- H - Water Sensitive Urban Design (WSUD)

* One-off minor development, where additional impervious area increase of building or structure is 50sqm or more.

** One-off minor development, where additional impervious area increase of building or structure is 40sqm or more

*** Geotechnical Investigation is required to support OSA design if more than 50sqm is to drain into the OSA

+ OSD is not required if all stormwater collected from impervious areas can be disposed by OSA

++ To drain basement or sub-floor areas only

Nuisance Overland Flooding to Low-Lying Habitable Floors and Basement Areas

- SWM11 Gutter Capacity - Where the proposed development will result in the exposure of low-lying habitable floors or basements including sub-floor areas, garages and low level car parking areas to potential ingress of surface runoff, the determination of the street gutter capacity based on the peak runoff for the 1% Annual Exceedance Probability (AEP) storm event will be required. This is to ensure that the likelihood of surface runoff from the street entering the proposed low-lying areas does not occur.
- SWM12 Flood Protection – For all low lying habitable floors or basements including sub-floor areas, garages and low level car parking areas, following determination of the 1%AEP design storm and gutter capacity, if it is found that there is potential for surface stormwater to enter into the property or minimum free boards cannot be achieved, it will be necessary to introduce a crest in the driveway ramp, or relocate the basement entry to a location where it is not vulnerable to water ingress, or raise floor levels to provide adequate free board protection.
- SWM13 Flood gate or barrier may be considered for the additions and alterations to existing properties, or a single dwelling house development if it is reasonably necessary and it can be demonstrated that the crest and driveway ramp gradient has met the maximum gradient requirement, as outlined in the relevant Australian Standard. For any other new development, (e.g. commercial or residential containing more than one dwelling etc.), the crest requirement (1% AEP + 0.5m freeboard) shall be addressed, and flood gate or barrier shall be used to meet the probable maximum flood (PMF) requirement.

Overland Flow and Flood Studies

- SWM14 Hydrological and Hydraulic studies - Where the proposal has the potential to obstruct natural flow paths, encroach over or adjacent to a public drainage infrastructure or a stormwater drainage easement, a driveway proposed into a basement, or sub-floor area, or the area proposed for development is potentially flood affected, an overland stormwater flow assessment or a flood study will be required when lodging a development application.
- SWM15 Design Storm - The design storm to be used to determine the flowrate for the overland flow assessment or a flood study shall be the peak 1%AEP storm event. A PMF storm event flowrate may also be required for flood risk management study. Design guidelines are referred to Section ***Stormwater Drainage Design***.

On-site Stormwater Detention Systems

General

- OSD1 Runoff Control - On-site Stormwater Detention (OSD) system is to be implemented to control the rate of runoff from development sites and subdivisions to limit and reduce the rate of runoff to the existing state of nature condition or better.
- OSD2 Alternative Methods – Combination of OSD with other runoff control can be considered depending on the type of development, site condition and the location of stormwater disposal.
- OSD3 Maintenance Schedule – where a development site requires to provide a OSD storage facility. A maintenance schedule shall be prepared by a suitably qualified engineer demonstrating a simple set out maintenance actions and the frequency of such actions including a sketch plan to ensure the constructed devices can be maintained by the future property owners and occupiers.

Exemptions

- OSD4 Exemptions - OSD will be applied to the developments types as listed in the table under the Section **Controls**. Exemption from OSD would only apply in the following situations:
- The proposal is a residential development consisting of only alterations and additions to an existing dwelling where the additional increase in impervious area is first on-off minor development and does not exceed 50sqm outside the footprint of the existing building and the land naturally falls towards the property street frontage with direct access to a formed public road/street.
 - The development consists of improvements only to the existing building such as a second floor extension or internal refurbishments, which are wholly confined within the footprint of the existing building. However, if the runoff from the gutter and proposed/existing downpipes are to be redirected to different sub-catchment area. OSD system requirement is still applied.
 - Site area is located within a 1%AEP flood affected area with the depth of flood water

greater than 150mm or subject to tidal influence. This does not include areas where it is affected by nuisance flooding caused by inadequate capacity of the drainage system or localised overland flow. The runoff from the development is directly discharged into one of the bays or waterways and does not pass through any public drainage system. Council should be consulted on this matter for further clarification.

- The runoff from the development is directly discharged into one of the bays or waterways and does not pass through any public drainage system (e.g. piped conduit, dish drain, open channels, kerb and gutter, public reserve, public roadway etc.) or that downstream properties could be adversely affected by the discharge from the development site.
- An alternative method of stormwater disposal is applied such as an on-site absorption system, which fully caters for the site's runoff.
- Council should be consulted on this matter for OSD exemptions for further clarification.

Site Storage Requirement and Permissible Site Discharge

- OSD5 Catchment Based Method - The Catchment Based Method for determining the Site Storage Requirement (SSR) and Permissible Site Discharge (PSD) is to be used to size OSD systems for all development types except demolition works. The hydrological Site Based Method (ILSAX or DRAINS) is only to be used if more than 30% of the site cannot be routed through the OSD system.
- OSD6 Design Parameters - OSD design parameters are as follows:
- Site Storage Requirement (SSR) is 200cum per hectare.
 - Permissible Site Discharge (PSD) is 180L/s per hectare.
- OSD7 Area for Calculating Volume and Discharge – For the Catchment Based Method, the area to be used in calculating the OSD volume (SSR) and discharge (PSD) shall be based on the total site area.

For the hydrological Site Based Method, the PSD shall be calculated on a 0%

pre-development site impervious area during the peak flow of overall critical storm for the 20% AEP storm event and the volume shall be sized on the basis of reducing the fully developed (i.e., controlled + uncontrolled area) site discharge for the peak flow of overall critical storm for the 1% AEP storm event back to the PSD.

OSD8 Alterations and Additions - In the case of development type 2, where the proposed alterations and additions are to be completely separated from the existing building/s and represent less than 10% of the total site area, then the SSR and PSD can be determined from the proposed footprint of the additions only. In the case if the existing dwelling is to be retained and adding a secondary dwelling with subdivision (e.g., Torren or Strata title) to become Dual Occupancy type (i.e., Type 4a) OSD requirement shall apply to the whole Lot. Including existing and propose development.

OSD9 Maximum Site Discharge - The maximum discharge from the site must not be greater than the calculated PSD. This would include both piped and uncontrolled flows. That is, the total outflow from the site must not exceed the PSD.

OSD10 Stormwater Discharge – Each development site shall generally have one (1) stormwater discharge connection point to the kerb. The maximum discharge to the kerb at any single point shall be limited to 25L/s per 15 lineal metre of frontage, for all storm events. Pressure stormwater drainage system directly to street kerb is prohibited.

Orifice controls

OSD11 Orifice Types - Orifice is to be a circular shaped hole drilled in a flat plate made from a stainless steel plate of 3mm thick or 5mm when the orifice diameter exceeds 150mm and 200mm x 200mm minimum dimension. The orifice shall be formed to a circular square edged cut (circular hole is to be pre-drilled into the plate to 0.5mm accuracy) and the plate permanently fixed to an oversized conduit or pipe stub if the hole is greater than 100mm.

OSD12 Permanent Fixture of Orifice - Dyna bolts or equivalent are to be used to permanently fix orifice plates. For orifice stubs/pipes, the fixture is to be such that tampering and removal is minimised.

OSD13 Minimum Orifice Diameter - For orifice plates, the size of the orifice shall not be less than 50mm in diameter to avoid blockages.

OSD14 Orifice equation - The orifice equation to be used is:
 $Q = C.A.\sqrt{(2.g.h)}$
 Where
 D = diameter of orifice = $\sqrt{(4.A/\pi)}$ in metres
 A = area of orifice hole in square metres
 $\pi = \text{pi} = 3.1416$ (dimensionless)
 g = acceleration due to gravity = 9.81 m/s²
 C = orifice coefficient (dimensionless). Ref table
 h = depth of water to centre of orifice in metres

OSD15 Orifice Coefficient - Orifice coefficients are given in table below.

Orifice Type	Discharge Coefficient
Stainless steel flat plate with circular hole	0.6
Pipe stub 100mm long	0.8

High Early Discharge (HED) Control

OSD16 HED Control Pit Type - The control pit in the OSD system must be a "High Early Discharge" (HED) arrangement for the Catchment Based Method. The following requirement shall be met:

- The discharge to the storage shall commence at a minimum of 75% of the PSD as calculated for the various storm frequencies.
- The overflow structure from the discharge control pit to the storage (weir, pipe, grate) shall convey the maximum 1% AEP flow to the discharge control pit less the initial HED through the orifice.
- A minimum of 85% of the area drainage into the OSD system shall be drained directly to the HED control pit or chamber.
- The volume of the discharge control pit shall be small in comparison to the volume of the storage.

OSD17 Non HED Control Pit Type - If a HED control configuration is not used, then an additional 20% volume must be added to the final or adjusted site storage require (SSR) volume for the Catchment Based Method.

Tail-water influence

OSD18 Orifice Control - The centre-line of the orifice must be higher than the water level at the point of connection into the existing receiving drainage system otherwise a "drowned orifice" or downstream control occurs.

OSD19 Drowned Orifice - Where a "drowned orifice" occurs, supporting hydraulic calculations will need to be submitted to ensure that there is no loss in storage volume as a result.

OSD20 Control Types - Adopted 1%AEP tailwater control levels at the point of connection are given in table below.

Discharge point	Water level to be adopted
Kerb and Gutter	At the top of kerb
Pipe outlet or culvert	Top of conduit if super-critical flow or "free outlet"
	Depth as calculated if sub-critical flow
Pit	150mm above top of pit
Open channel	Depth as calculated
Waterway or Bay	Depth at High Tide

Site area to be routed through OSD

OSD21 Area to be routed through OSD System - The majority of the site runoff must be routed through the OSD facility.

A maximum of 30% of the site is permitted to bypass the OSD system if the Catchment Based Method is to be used. The 30% bypass must not be made up of more than 50% of the site's impervious areas.

For the Site Based Method, a maximum of 50% of the site is permitted to bypass the OSD system subject to the overall permissible site discharge achieving Clause OSD7 requirement.

OSD22 Adjustment of Volume and PSD due to Bypass for the Catchment Based Method.

For a bypass of 30% of total site area, an additional 20% increase in the calculated SSR volume shall be applied and the PSD shall be reduced by 20%. Proportional increase of the basic volume and reduction of the PSD shall apply between 0% and 30% bypass.

Trash screens

OSD23 Purpose - A rustproof screen or cage is to be used to protect the outlet from potential blockages.

OSD24 Removable - The screen or cage must be removable for ease of maintenance and inspections.

OSD25 Material - The screen or cage is to be made from Lysaght's maxi-mesh RH3030 or similar material.

OSD26 Minimum Surface Area - The minimum surface area of the screen is to be 50 times the area of the orifice outlet.

OSD27 Location - The screen or cage shall completely protect the outlet. It is to be located at a distance 1.5 times the orifice diameter or 200mm away, whichever is the greater.

OSD28 Orientation - Where possible, the main incoming line is to flow across the face of the mesh.

OSD29 Lifting Handle - The screen or cage shall include a lifting handle for ease of removal for inspection and maintenance.

Minimum requirements for underground OSD systems

OSD30 Purpose - Underground OSD systems shall be designed and located to attenuate and safely control stormwater runoff from the site to the public road gutter or receiving drainage system.

OSD31 Overland Flow path - Where likely overflow from the OSD facilities pass through private property and a controlled and safe overland flow path cannot be provided, the overflow is to be fully contained within an underground piped drainage system with a design capacity equivalent to the peak 100-year ARI storm event.

OSD32 Location - Underground OSD systems must be located external to all building footprints. This includes basements, ground floor parking areas, garages, or patios unless all access points and emergency overflow provisions are external to the structure.

Underground OSD systems must be located outside the root zones of trees that are to be retained.

Underground OSD systems must be located in areas which will not impede existing overland flow paths, or cause concentration or diversion of stormwater into public or private property/s resulting in the undermining of existing structures or services or jeopardising public safety or cause public nuisance.

Underground OSD systems must not be located within any deep soil areas or within the front building setback. OSD systems must not restrict the provision of required deep soil or landscape areas as prescribed in accordance with this DCP.

Where underground OSD systems are located under landscaped areas, a minimum topsoil cover of not less than 800mm over the tank should be provided.

Underground OSD systems are to be located where they can be easily accessible for inspections and routine cleaning.

In this regard, for development types 4a, 5, 6 and 7, the following requirements must be met:

- The underground OSD facility must be located in common areas and not in private courtyards
- The location of the underground OSD system is such as to ensure that the facility is located where easy unimpeded access is possible for routine inspections and maintenance requirements. It needs to be accessible for third party inspectors including Council

OSD33 Provision of Formal Surcharge Path - A formal surcharge path must be provided even when the 1%AEP flows can be fully catered for by the system, to account for pipe blockages and higher intensity storms.

OSD34 Easements - Easements must be created over private property/s for pipe systems and surcharge paths through adjoining property/s.

OSD35 Structural Adequacy - Underground OSD systems are to be designed to be structurally sound and able to adequately withstand all service loads.

OSD36 Adequate Soundproofing - Underground OSD systems must be adequately soundproofed to minimise noise when stormwater is collected or discharged.

OSD37 Drainage of Base - The base of the OSD system must be graded to drain completely. Permanent water ponding in the tank encourages insect infestation and will not be acceptable. The base of the structure is to have a minimum longitudinal gradient of 0.7% to the outlet.

OSD38 Material - Underground OSD systems must be constructed from reinforced concrete, pre-fabricated material or proprietary system/s approved by Council. Atlantis units or similar type systems are not to be used for detention storage due to reduced storage and difficulties with maintenance and inspections.

OSD39 Inspection Access - Underground OSD systems must have at least one (1) inspection access opening over the storage outlet and/or discharge control outlet. This inspection point must be a minimum 600mm x 600mm in dimension subject to the internal depth of the pit referring to the latest AS3500.3.

For ease of maintenance at least one (1) additional access at the extreme corner of the tank must be provided.

Additional access or flushing points shall be provided for irregular shaped structures to allow for easier access for routine cleaning and maintenance.

Underground OSD systems must have additional access points at distances of not less than 5000mm.

OSD40 Confined Spaces - Underground OSD systems must comply with **AS2865 Safe Working in a Confined Space** and be appropriately designed to eliminate gas build-up.

OSD41 Steps - Underground OSD systems must have step irons when the depth exceeds 1200mm.

Minimum requirements for above-ground OSD systems in landscaped areas

OSD42 General - Above-ground OSD systems include areas where detention storage is provided in soft and hard landscaped area and above ground storage structures such as rainwater tanks.

OSD43 Purpose - As with Underground OSD systems, above-ground OSD systems are to be designed to collect, attenuate and safely control all stormwater runoff from the site to the public road gutter or drainage system.

OSD44 Overland Flow path - Where likely overflow from these OSD facilities is across private property and a suitable overflow path cannot be provided then the overflow shall be contained within an underground piped drainage system with a design capacity equivalent to the peak 1%AEP storm event. This underground system (both pits and pipes) shall be designed to receive and fully contain controlled flows and overflows from the OSD facility.

OSD45 Location - Above-ground OSD systems must be located external to all building footprints, basements, ground floor parking areas, garages, or patios unless access points and emergency overflow provisions are either provided externally or are easy to get to for the purpose of inspections and maintenance to the system.
Above-ground OSD systems must not be located across the boundary/s of allotment/s. Above-ground OSD systems must not be located forward of the front building line. Above-ground OSD systems must not be located such as to restrict pedestrian access from the public road to the building. Where above-ground structures such as rainwater tanks are to be used, they must be located in an area least visually obtrusive and in compliance with Councils other related planning requirements.

Above-ground OSD systems must be located where they can be easily accessible for inspections and routine cleaning. In this regard, for development types 4a, 5, 6 and 7, the following requirements must be met:

- The OSD facility must be located in common areas and not in private courtyards
- The location of the OSD system must ensure that the facility is located where easy unimpeded access is possible for routine inspections and maintenance requirements.

OSD46 Provision of Formal Surcharge Path - A formal surcharge path must be provided even when the 1%AEP design flow is fully accommodated within the piped system to account for greater storm intensities and blockages.

OSD47 Easements - Formal easements must be created on private property for pipe systems and surcharge paths.

OSD48 Adequate Soundproofing - Above-ground OSD systems must be adequately soundproofed to minimise noise when stormwater is collected or discharged.

OSD49 Adequate Waterproofing - The barrier/retaining wall around the perimeter of the OSD storage area must be constructed of watertight concrete/masonry/brick type material or a fully waterproof (prefabricated) material or equivalent. Timber construction is not permitted.

OSD50 Accessibility of Control Structures - Control structures must be located where they can be easily accessible for routine inspections and cleaning.

OSD51 Drainage of OSD Base - The base of above-ground OSD systems must be graded to drain completely. In soft landscaped areas, the gradient of the base must not be less than 1.0%.

OSD52 Subsoil Drainages - To avoid ground saturation in soft landscaped areas, subsoil drains shall be installed where gradients are less than 1.0%.

OSD53 Good Design - Above-ground OSD systems must be designed in a manner which minimises inconvenience, unsightliness and hazard and is compatible with the proposed development in terms of functionality and purpose.

OSD54 No floatable material such as woodchip, leaf mulch, pine bark will be permitted for planting within OSD basin and any inundation area that causing debris and stormwater pollution. Grass surface or stone gravels (e.g., 30mm or greater rock mulch) will be permitted.

OSD55 Maximum Ponding Depths - Above-ground OSD systems must not allow water to pond more than 200mm in tennis court areas and other hard paved surface areas.

Above-ground OSD systems in landscape area shall be limited to pond no more than 300mm. Ponding greater than 300mm may be considerable if the following parameters are satisfied:

- Restricted access is enforced but possible to walk out from the ponding area.
- Embankment batter slopes are greater than 1 in 4.
- OSD basin is fully fenced off with approved pool fencing with child safety lock by the principal certifier, self-closing gates with batter slopes into the basin, around its perimeter, not to exceed 1 in 6.
- The maximum depth of ponding in landscaped areas must not exceed 1200mm, regardless of other restrictive measures implemented.
- For location where the vertical drop is greater than 900mm and where the rock batters are steeper than 1H:1V, a safety fence must be installed to prevent falls.

Minimum requirements for above-ground OSD systems in driveways and car parking areas

OSD56 Purpose - These OSD systems must be designed and located to collect, attenuate and safely control stormwater runoff from the site to the public road gutter or drainage system.

OSD57 Overland Flow path - Where likely overflow from these OSD facilities is across private property and a suitable overflow path cannot be provided, the overflow shall be contained within an underground piped drainage system with a design capacity equivalent to the peak 1%AEP storm event. This underground system (both pits and pipes) shall be designed to receive and fully contain controlled flows and overflows from the OSD facility.

OSD58 Location - These above-ground OSD systems must be located external to all building footprints. This includes basements, ground floor parking areas, garages, or patios unless all access points and emergency overflow provisions are external to the structure.

These above-ground OSD systems must not be located such as to restrict pedestrian access from the public road to the building.

Control structures must be located where they can be easily accessible for routine inspections and cleaning.

OSD59 Provision of Formal Surcharge Path - A formal surcharge path must be provided even where the 1%AEP design flow is fully accommodated within the piped system to cater for greater storm intensities and blockages.

OSD60 Drainage of Base - To avoid localised ponding within the detention area, the base of the OSD system must be graded to drain completely, with gradients not less than 0.5% in concrete paved areas or similar and 0.7% in bitumen surfaced areas.

OSD61 Good Design - These above-ground OSD systems must be designed in a manner which minimises inconvenience, unsightliness and hazard and is compatible with the proposed development in terms of functionality and purpose.

OSD62 Adequate Waterproofing - These above-ground OSD systems must be totally impermeable unless permeable pavement has been designed as a function provided any water penetration will not affect adjoining buildings, structures and the like.

OSD63 Maximum Ponding Depths - Water is not permitted to pond more than 200mm in depth for all storm intensities, in these above-ground OSD systems.

Overland Flows and Flow Paths

OSD64 Existing and Natural Overland Flow paths- Existing and natural overland flow paths for surface stormwater are to be maintained whenever possible.

Overland flows or surface runoff collected from the site or passing through the site from upstream property/s is not permitted to be concentrated and channelled onto adjoining property/s.

OSD65 Concentration of Overland Flow paths - Where surface runoff or overland flows become concentrated, they will need to be safely re-directed to the street or to the site's OSD facility.

OSD66 Diversion of Surface Runoff or Overland Flows to OSD System - Where concentrated surface runoff or overland flows are to be controlled through the OSD facility, the OSD system must be designed to cater for any additional inflow.

OSD67 Emergency Overland Flow path - To cater for unexpected blockages in the OSD system or additional runoff in excess of the design storm, provision must be made for an emergency flow path from the OSD system to the street such as installing overflow weirs or spillways. The width of overflow weirs shall not exceed more than 2000mm across any property frontage.

OSD68 Designated Flow paths - Any likely overflow from these facilities must be fully contained within designated flow paths and must be conveyed safely to the nearest public road reserve or waterway.

Designated flow paths shall be designed for the peak 1% AEP storm event. Open channels, kerbs, pits and pipes may be used to contain runoff within the flow path.

Freeboard

OSD69 Definition - Freeboard refers to the clearance distance between the maximum water level and the surface level of habitable floors and garage floors.

OSD70 Adequate Freeboard - Adequate freeboard or clearance distance between the maximum depth of ponding in the detention system and the finished floor level of all habitable areas, garages, storage facilities, etc., are to be provided so that damage to goods and materials, nuisance flooding, or hazard is avoided.

OSD71 Minimum Freeboard - The minimum freeboard requirements for OSD pondage are tabled below.

Maximum water level in (A)	Finished floor level (B)	Minimum Freeboard between (A) and (B)
Detention facility	Warehouse	150mm
Detention facility	Factory	150mm
Detention facility	Garage	150mm
Detention facility	Carport	150mm
Detention facility	Office space	300mm
Detention facility	Habitable rooms	300mm
Detention facility	Living rooms	300mm
Detention facility	Retail space	300mm
Detention facility	Store rooms	300mm
Detention facility	Show rooms	300mm

***Important Note:** Refer to Clause **SW23** and the Flooding Control Section of Council's **Development Control Plan** for freeboard requirements in flood prone land.

All proposed developments are generally prohibited in known flood areas, flood ways and flood zones unless it can be demonstrated that a fail safe emergency evacuation route can be provided for occupants and that the proposed development will not cause an increase of the flood waters both upstream and downstream of the proposed development.

Council's Stormwater Policy is for the removal of such flood affected developments from known flood ways and the prohibition of future developments in such floodways, wherever possible, to protect life and amenity.

Discharge Points

OSD72 Maximum Concentrated Stormwater Runoff

- The maximum allowable discharge of concentrated stormwater runoff to the kerb and gutter at any single point is 25L/s. This limit applies even if the permissible site discharge exceeds this rate.

If the discharge needs to exceed 25L/s, the following alternatives may be considered:

- Alternative 1 - Discharge made directly to the nearest Council piped drainage system, or to an approved piped drainage system or lay a new pipe and pit system to connect into the existing drainage infrastructure. This new line must be designed and installed to Council and the relevant road authority standards and become a public system. Note: Consent under Section 138 of the Roads Act 1993 will be required prior to commencement of roadwork.
- Alternative 2 - Increase the size of the detention system (if on-site stormwater detention is required) to limit the outflow to the street kerb to 25 L/s. This alternative is not permitted in specific zoning areas (MU1, SP4, R1, R4, E4),
- Alternative 3 - If there is enough property frontage width, allow split flows to the kerb at a minimum distance of 15 metres apart between kerb outlets. This alternative is not permitted in specific zoning areas (MU1, SP4, R1, R4, E4),
- Alternative 4 - Discharge directly into a watercourse or channel or to a stormwater channel or to the Bay, subject to joint approval from Council and other relevant governing authority. Adequate protection against scour and erosion at the point of discharge must be provided

OSD73 Maximum Discharge Velocity - The maximum discharge velocity into an unlined, that is, other than concrete channel shall be in accordance with the Department of Sustainable Natural Resources document **Managing Urban Stormwater, Soils and Construction**, table 5.1 maximum discharge flow velocities in waterways.

OSD74 Approval for Connection into Existing Council Piped Drainage System - Connection into an existing Council piped drainage system may be permitted subject to Council approval.

OSD75 Construction of New Piped System - If direct connection outside of the property frontage cannot be obtained, this will require the construction of new inlet pits and the laying of a new pipeline, to Council standards, to connect into the existing line. Minimum Council design standards are given in Section under **Stormwater Drainage Systems**.

OSD76 Minimum Drainage Infrastructure - Council encourages minimising the number of pits, junctions and pipe lengths in the road reserve. However, pits are preferable over bend joins to eliminate blind junctions such that ease of maintenance can be achieved.

OSD77 Private Stormwater Drainage System - Any private stormwater pipe leaving the site, must be laid and connect to the discharge point within the subject site frontage.

An angle of not less than 45 degrees to the front boundary line is recommended. If the pipe needs to be laid at a more acute angle or must run parallel with the kerb line in order to achieve gravity discharge to the kerb or connection into an under-ground system, then it must be taken to the kerb at 45 degrees and then a minimum 375-dia RCP is to be laid parallel to the kerb line connecting into the nearest downstream underground public drainage system. Standard gully inlet pits with minimum 1.2 metre clear opening long kerb lintels (subject to required inlet capacities) will need to be constructed in front of the subject site frontage and at all bends, junctions and minimum distances between pits as described in **Australian Standard AS3500.3 and Australian Rainfall and Runoff**.

OSD78 Equivalent Conduit Sizes - Conduits laid to the kerb face must be sized to cope with the design flow and with at least 50mm cover. Where cover is inadequate, the following pipe equivalencies shall be used:

- 100mm-dia equivalent to one (1) 125mm x 75mm x 5mm thick RHS
- 150mm-dia equivalent to one (1) 200mm x 100mm x 6mm thick RHS

OSD79 Outlets through Sandstone Kerbing - Outlets laid through existing sandstone kerbing will require drilling through sandstone kerb subject to Council approval.

OSD80 Pipe Class - The pipe class must be adequate to withstand proposed traffic loads, street tree loading and prevent ingress of street tree roots.

OSD81 Rectangular Hollow Sections - All Rectangular Hollow Sections (RHS) are to be hot-dipped galvanised or stainless steel material.

Discharge to Different Catchment

OSD82 Gravity Drainage - Stormwater is to be drained in the same direction as the natural fall of the land by gravity means.

OSD83 Diversion of Stormwater to Different Catchment - Diverting stormwater runoff from one catchment (or sub-catchment) to another catchment (or sub-catchment) is generally prohibited.

OSD84 Consent for Stormwater Drainage to Different Catchment will only be considered by Council, provided that all conditions listed below are achieved to Council's requirements:

- Adjacent downstream property owner/s have indicated that they are not prepared to grant easement/s to permit the drainage of the low lying property to follow the natural fall of the land. Proof of genuine attempt at easement acquisition in the form of correspondence will be required.
- A detailed hydrological and hydraulic analysis and calculation in terms of receiving drain capacity analysis and clearly demonstrating that the redirection flow will not cause any adverse third part impact along the diverted route outside the street network. Also, it can be demonstrated that there are no existing drainage problems downstream in the catchment where the runoff is to be directed.
- Discharge made directly to the nearest consented piped drainage system from the road authority within the subject site frontage or lay a new pipe and pit system to connect into the existing drainage infrastructure downstream. The new line must be designed and installed to Council's standards or the relevant road authority standard and will become a public system. The capacity of the receiving underground stormwater drainage system must be demonstrated by calculation and comprehensive analysis and be able to determinate that the system is sufficient to accommodate the additional flow or redirect flow from the development. The calculation and analysis shall be prepared and certified by a suitably qualified engineer with a National Engineering Register (NER) Accreditation.

Note: Consent under Section 68 (Part B) of the Local Government Act 1993 from Council and Section 138 of the Roads Act 1993 from the road authority must be obtained prior to obtaining consent for the development and activity.

OSD85 Drainage "Against Fall of Land" - Where drainage "against the fall of the land" is permitted, On-site stormwater Detention will be required, regardless of the development type. The volume and discharge is to be calculated on the areas (proposed/existing) being directed to the outlet.

OSD86 Low-Lying Property Drainage - For low lying properties, on-site stormwater absorption (OSA) may be used for the disposal of runoff from on-ground paved or roof areas that are unable to be collected and taken to the street frontage.

Legal Obligations

OSD87 Positive Covenant and Restriction on the Use of Land – All types of development which requires an on-site stormwater quantity and/or quality management system (on-site stormwater detention, on-site stormwater absorption, stormwater quality improvement device and/or mechanical pump-out system) will require a Positive Covenant and Restriction on the use of land, in favour of the City of Canada Bay Council on the Title.

OSD88 Purpose of Positive Covenant - The purpose of the Covenant is to ensure that the registered proprietor of the land is made aware and takes responsibility for the control, care and maintenance of the constructed on-site stormwater management system.

OSD89 The Purpose of Restriction on the Use of Land - The purpose of the Restriction ensures that the system cannot be altered in any manner, shape or form.

OSD90 Newly Created Allotments - For newly created parcels of land, these terms shall be created under Section 88B of the Conveyancing Act 1919.

OSD91 Existing Land - For existing titles, the terms of Positive Covenant and Restriction on the use of land shall be created by an application to the Land Titles Office using Forms 13PC and 13RPA.

OSD92 Standard Wording - Standard wording for Positive Covenant and Restriction on the use of land are given in the Appendix.

On-site Stormwater Absorption Systems

General

OSA1 Application - On-site stormwater absorption (OSA) systems may be used for disposal of the site's stormwater runoff. Refer to the Section under **Controls** for when it can be applied.

OSA systems can be used to dispose of stormwater on the site subject to the suitability of soil conditions and site constraints. Typical OSA systems include rubble drains, rubble trenches, and on-ground infiltration systems.

OSA can be used as an alternative method of stormwater disposal for 'low lying' properties where an easement through adjoining downstream properties cannot be obtained.

Design Principles

OSA2 Standard Design Methodology - The OSA system shall be appropriately sized, based on the area to be captured, using the methodology as given in the following table:

Total Impervious Area to be captured (sqm)	OSA Design Methodology*
≤ 50	Standard trench detail - 'Jumbo 410' type or equivalent material (refer to Appendix). The trench length is to be 4 metres for every 25sqm of catchment (hard stand, roof) area draining to it.
>50	Geotechnical Report to be prepared by a Geotechnical Engineer. Trench area and volume calculated from soil infiltration rate and Report recommendations. Mass Curve Method can be used.

***Important Note:** Design Constraints apply

OSA3 Location - OSA systems are to be located in 'soft' landscape areas, such as in the garden and vegetated on-ground areas. In 'hard' landscape areas, the use of porous pavement (pervious paving) is preferred.

Absorption systems shall not be located under or over any sewer service without Sydney Water approval.

Absorption systems are not to be located in rock (most non-sedimentary rocks and some sedimentary rocks such as shale) which has zero or near-zero permeability. Where rock is encountered, the base of any proposed gravel filled trench shall be set at least 500mm above the rock.

OSA systems are not permitted in shallow soil over bedrock.

Absorption trenches must follow the line of contours.

OSA4 Access - For routine inspections and maintenance, an access chamber is to be provided at either end of the OSA system.

OSA5 Water Sensitive Urban Design (WSUD) - OSA systems may be used in combination with WSUD systems and integrated as part of its performance principals.

OSA6 Design Storm Event - Runoff generated from the development must be fully catered for by the absorption system, for all storm events up to and including the 1%AEP storm event for all duration from 6 minutes to 72 hours inclusive.

OSA7 Soil Type - Soils to be used for OSA systems **must not** be predominantly loose aeolian sands or clay soils.

Suitable soils must have a uniform thickness of at least 3.0 metres.

A soil assessment and permeability test is required from a qualified Geotechnical Engineer where runoff is collected from an area greater than 50sqm.

Suitable soils must have hydraulic conductivity values greater than 1×10^{-6} m/s and the minimum absorption rate to be adopted shall be limited to 1.0 L/sqm per second irrespective of the absorption rate achieved by any geotechnical testing.

- OSA8 Minimum Clearance from Structures and Buildings - Clearance distance between all buildings, footings, structures and downstream boundaries are to be at least 3.0 metres from the proposed OSA system.

Minimum clearances are not to be reduced without a supporting Geotechnical Report.

The minimum clearance distance between sewer mains and the proposed absorption system is to be at least 1.0 metre unless approved by Sydney Water. Where clearance between buildings or structures to the OSA system cannot be provided, the structure is to be supported on a pier and beam system to a minimum 300mm below the base of the OSA system, which must be certified by a Structural Engineer.

- OSA9 Steep Sites - Absorption systems will not be permitted on 'steep' sites with slopes greater than 5%, unless there is a supporting Geotechnical Report.
- OSA10 High Water Table - Absorption systems will not be permitted where there is the presence of a high water table or the location of the OSA system would likely cause the water table to rise.
- Where a high water table is encountered, the base of any proposed gravel filled trench shall be set at least 500mm above the water table.
- OSA11 Sediment Traps and Trash Screens - Runoff directed into the OSA system must pass through an approved silt trap and filter system to remove all debris, silts, sands, etc. prior to absorption.
- OSA12 Dispersion System - To ensure any overflows from the OSA system are not concentrated onto adjoining downstream properties, a dispersion system shall be provided at the overflow outlet.
- OSA13 Maintenance Schedule – where the OSA is permitted in the development site. A maintenance schedule shall be prepared by the qualified suitable engineer demonstrating a simple set out maintenance actions and the frequency of such actions including a sketch plan to ensure the constructed devices can be maintained by the future property owners and occupiers.

Charged Systems

General

- CH1 Application - 'Charged Systems' refers to a network of sealed or watertight stormwater drainage pipes which convey stormwater under constant pressure, requiring sufficient hydraulic head to move water.

Where a site naturally falls to the rear and the property is not benefited by any stormwater drainage easement and following unsuccessful attempts at obtaining a drainage easement and other drainage disposal methods such as OSA, have been found to be inappropriate for the site, then charged systems may be considered, to convey stormwater to the street frontage. Refer to Section **Controls**, for when it can be applied.

Charged systems are generally not to be used if a gravity feed system is available to dispose of stormwater off the site to the nearest downstream Council drainage system.

Important Note: Charged systems are only permitted when used in combination with rainwater re-use facilities.

- CH2 Permissibility - Charged systems may only be used if address the requirements listed in Clause OSD84 under On-Site Stormwater Detention Systems.
- Genuine attempt at easement acquisition has been undertaken and the downstream property owner/s have indicated that they are not prepared to grant easement/s to permit the drainage of the property to follow the natural fall of the land - written evidence is to be submitted as proof
 - The soil absorption characteristics and other physical constraints indicate that an OSA system is not feasible to cater for the development. A Geotechnical Report, indicating that the soil has very poor absorption rate, may be submitted as proof
 - A detailed hydrological and hydraulic analysis and calculation in terms of receiving drain capacity analysis and clearly demonstrating that the redirection flow will not cause any adverse third part impact along the diverted route outside the street network. Also, it can

be demonstrated that there are no existing drainage problems downstream in the catchment where the runoff is to be directed.

- Discharge made directly to the nearest consented piped drainage system from the road authority within the subject site frontage or lay a new pipe and pit system to connect into the existing drainage infrastructure downstream. The new line must be designed and installed to Council's standards or the relevant road authority standard and will become a public system.

Note: Consent under Section 68 (Part B) of the Local Government Act 1993 from Council and Section 138 of the Roads Act 1993 from the road authority must be obtained.

Design Principles

- CH3 Permissible Site Discharge - The rate of runoff discharged to the street must not exceed the overall Permissible Site Discharge (PSD) calculated from Section **On-site Stormwater Detention Systems**.
- CH4 Combination of OSD, OSA and Charged Systems - The design of a charged system may be used in combination with an OSD and/or OSA system to minimise the runoff to the street.
- CH5 Watertight System - Charged lines may be taken directly from the roof gutter to the street and must be fully watertight.
- CH6 Rainwater Re-use Overflows - Overflows from the rainwater re-use system are to be drained to the external on-site stormwater drainage system, OSD or OSA system if these have been designed to accommodate the runoff.
- CH7 Outflows from OSD systems under pressure - Outflows from OSD systems are generally to be by gravity feed ('free outlet' control). However, 'drowned outlets' may be permitted if supporting calculations can be provided showing that sufficient pressure head can be generated to direct flows to the street via a charged system and that the permissible discharge for the site can be maintained. A suitable hydrological program such as DRAINS can be used to model such situations.

- CH8 Design Storm Event - The charged drainage system including roof gutters pipes and pits is to be designed for the 5 minutes duration in 1%AEP storm event.
- CH9 Minimum Height of Pressure Head - The minimum height difference between the roof gutter level and the discharge pit must be at least 2.0 metres unless supporting calculations showing a hydraulic grade line for the 1%AEP storm event indicates that the system can drain to the street with a 300mm free board to the gutter line.
- CH10 Gravity Feed to Street - The charged line must discharge to a sump pit within the property boundary and then gravity fed to the street underground drainage system.
- CH11 Location of Surge Pits - Surge pits are to be located such that any likely overflow is safely directed to the street or away from structures and buildings.
- CH12 Minimum Pipe Sizes for Pressure Application - Pipes must be minimum 150mm-dia in size for pressure applications and solvent welded unless hydraulic calculations support that a smaller pipe can be used. Pipes less than 80mm-dia will not be acceptable.
- CH13 Cleaning Eyes - Cleaning eyes within a sump must be installed at the lowest point in the system.
- CH14 Gutter Guards - Gutter guards are to be installed to minimise debris entering the charged system.

Mechanical Pump-out Systems

General

MP1 Application - Mechanical pumps are to be used generally to convey sub-surface, or seepage water in a basement (sub-floor) area or to drain minor surface runoff collected from weather exposed areas (e.g. vehicular ramp) of less than 50m² or 100m² for development Types 5, 6 and 7.

MP2 Pumping of Stormwater - The pumping out of stormwater runoff is only permitted if it addresses the requirements listed in Clause OSD84 under On-Site Stormwater Detention Systems, and:

- If consent is achieved in accordance with both the Roads Act 1993 and the Local Government Act 1993.
- Gravity drainage cannot be achieved.
- Genuine attempt at easement acquisition has been undertaken and the downstream property owner/s have indicated that they are not prepared to grant easement/s to permit the drainage of the property to follow the natural fall of the land - written evidence is to be submitted as proof.
- An OSA system cannot be installed because it does not satisfy OSA Design Principles.
- There is no possibility of nuisance or flooding or damage to adjoining buildings and structures in the event of pump failure. The pump-out system shall be located at least 3 metres away from the downstream boundaries.
- A detailed hydrological and hydraulic analysis and calculation in terms of receiving drain capacity analysis and clearly demonstrating that the redirection flow will not cause any adverse third part impact along the diverted route outside the street network. Also, it can be demonstrated that there are no existing drainage problems downstream in the catchment where the runoff is to be directed.

- Discharge made directly to the nearest consented piped drainage system from the road authority within the subject site frontage or lay a new pipe and pit system to connect into the existing drainage infrastructure downstream. The new line must be designed and installed to Council's standards or the relevant road authority standard and will become a public system. The capacity of the receiving underground stormwater drainage system must be demonstrated by calculation and comprehensive analysis and be able to determinate that the system is sufficient to accommodate the additional flow or redirect flow from the development. The calculation and analysis shall be prepared and certified by a suitably qualified engineer with a National Engineering Register (NER) Accreditation.

Note: Consent under Section 68 (Part B) of the Local Government Act 1993 from Council and Section 138 of the Roads Act 1993 from the road authority must be obtained prior to obtaining consent for the development and activity.

MP3 Disposal of Sub-surface Water – Because it is a relatively constant source that promotes algal growth and constant wetness, draining of sub-surface water, whether by pumping or by gravity; either directly or indirectly, to the kerb is not permitted except by way of a controlled release specifically and separately approved under Section 138.1(d) of the Roads Act 1993. Generally, it is preferable to connect such sources to the nearest public underground drainage system. If that is not feasible and a thorough investigation has exhausted all acceptable alternatives then Council will consider an application for controlled release to the kerb.

Alternative methods of seepage water disposal and controlled release parameters are given in Section **Subsurface Water**.

Design Principles

MP4 Minimum requirements for Mechanical Pump-out Systems - The minimum requirements for mechanical pump-out systems for stormwater apply as follows:

- The pump-out system shall consist of two (2) mechanical pumps, connected in parallel, with each pump being capable of emptying the holding tank at a rate equal to the peak 1%AEP, 5 minute duration storm event. Any subsurface inflow rate during wet weather determined by an approved test procedure or estimated value is to be included.
- The mechanical pump must be capable of draining the surface runoff collected from weather exposed areas of not less than 50m² or 100m² for Types 5, 6 and 7.
- The capacity of the holding tank shall be calculated as above the level at which all pumps are automatically brought into operation.
- The minimum capacity (volume) of the holding tank well shall be adequately sized in accordance with **AS/NZS3500.3.2 - 1998, National Plumbing and Drainage, Part 3.2: Stormwater drainage - acceptable solutions**, but based on the 1%AEP, 5 hours design storm.
- A silt trap shall be provided on the inlet side of the holding tank.
- The rising main from the pump system shall be designed and installed in accordance with the pump manufacturer's specification.
- The rising main from the pump system shall discharge to a stilling sump within the property boundary and then gravity fed to the underground receiving drainage system or to an OSD system if the OSD system is piped to an underground receiving drainage system.
- A one-way valve is to be installed on the rising main outlet. The stilling sump is to be located such that any likely overflow is safely directed to the street and away from adjoining buildings and structures.

- The switching of the pumps shall be arranged so that they operate alternately.
- The pumps shall be provided with automatic level switches so that they operate simultaneously should the capacity of the tank be exceeded.
- An automatic alarm system shall be provided to warn of failure of any part of the pump system.
- The alarm shall have visual indicators and an audible alarm siren.
- A rechargeable battery back-up system for the alarm is to be provided in the event of power failure.
- Maintenance Schedule – where the pump-out system is permitted in the development site. A maintenance schedule shall be prepared by the qualified suitable engineer demonstrating a simple set out maintenance actions and the frequency of such actions including a sketch plan to ensure the constructed devices can be maintained by the future property owners and occupiers.

Legal Obligations

MP5 Positive Covenant and Restriction on the Use of Land - Mechanical pump-out systems regardless of whether or not used as OSD systems or subsurface water shall require the creation of a Positive Covenant and Restriction on the Use of Land affixed to the title requiring the owner of the property to maintain the system, not tamper with the system and indemnifying Council against liability in respect of any damage sustained as a result of the failure of the pump system and damage to neighbouring property or any other cause not in Council's control, and providing for regular inspection by the proprietor. Refer to Section **On-site Stormwater Detention Systems**, Section under **Legal Obligations**.

Scouring, Erosion and Water Quality Control

General

- SC1 Application - Where discharge of stormwater is to a natural watercourse, into bushland/ reserve, property or water way, adequate controls are required to protect these adjoining properties, bushland, roadways and receiving waters from erosion and degradation due to pollution, silt laden stormwater runoff and/or high concentration of runoff.
- SC2 Scour and Erosion Control Devices - Notwithstanding other stormwater controls that may be required such as OSD (flow attenuating devices) or OSA, appropriate scour and erosion control devices are to be installed at the outflow to the system.

Minimum Requirements

- SC3 Application - The installation of scour and erosion control devices and other stormwater management measures are required as follows:
- Appropriate scour protection devices installed at all outlets to stormwater conduits
 - Additional installation of pollution control devices at the source, on-line*, off-line or at the end of the line to control sediment laden overland stormwater flows
 - Stormwater management measures shall not be located on-line in water courses or within riparian zones or areas of remnant native vegetation
 - Where the volume of runoff is considered high, it may be appropriate to include an OSD (for sites larger than 1200 sqm) and/or OSA system/s to reduce runoff into the receiving waterway
 - The type of available control or treatment required for each development type that drains directly into a natural watercourse, bushland/ reserve, or water way are tabled below:

Development Type	Control Type
1	E*
2, 3, 4, 4a, 6, 8	A, B, E*, F, H
5, 7	A, B, E*, F, G*, H

Key:

Development Type

- 1 - Demolition only
- 2 - Alterations and additions
- 3 - New dwelling with existing footpath along frontage
- 4 - New dwelling with no existing footpath
- 4a - Dual occupancy, town homes etc.
- 5 - Residential Flat Building (RFB), commercial and mixed developments
- 6 - Development under SEPP (Housing) 2021
- 7 - Substantial development
- 8 - Subdivision of one into two allotments

Water Quality Treatment System

- A - On-site Stormwater Detention System (OSD)
- B - On-site Retention System or Absorption System (OSA)
- C - "Charged" System
- D - Mechanical Pump-out System
- E - Scour and Erosion Control System/Devices
- F - Rainwater Re-use System
- G - Sediment Pond/Water Quality Device
- H - Water Sensitive Urban Design (WSUD)

* denotes mandatory requirement

- SC4 Pollutant Load to be Retained - Water quality treatment systems or pollution control devices such as sediment ponds where required to be installed are to ensure the following pollutant loads are retained on the site:

Pollutant Type	Percentage Retention of Post-development Loads
Total suspended solids (TSS)	80%
Total phosphorus (TP)	45%
Total Nitrogen (TN)	45%
Gross Litter	All Litter - 70% Material (>50mm) - 70%

- SC5 Monitoring Water Quality - Water quality treatment systems or pollution control devices are to be installed and monitored regularly to ensure that they achieve their treatment objectives and that their performance meets the above criteria. If they fail to meet these targets, or if so required by the Appropriate Regulatory Authority (ARA), it shall be modified to achieve them and the system upgraded.
- SC6 Ease of Maintenance - Water quality treatment systems or pollution control devices are to be designed to ensure ease of maintenance.
- SC7 Scour Protection - Scour protection devices shall include embankment stabilisation e.g. rock walls, concrete aprons, gabions, turfing, jute mesh, energy dissipating units, or other more appropriate erosion control devices approved by Council. Preference is for 'soft engineering' solutions.
- SC8 Control Devices near Creeks - Stormwater pollution control devices may not be appropriate within or adjacent to creek locations. In this regard, The Department of Planning and NSW Fisheries shall be contacted for advice about suitable erosion control measures here.
- SC9 First Flush - Stormwater pollution control devices to remove pollutants during the 'first flush', shall be installed within the site. These include proprietary items such as gross pollutant trap, hydrodynamic separation, filters, basket or silt and grease arrestors. Installation of these devices shall be in accordance with the manufacturer's specification.

- SC10 Large Scale Control Devices - Other sediment control devices such as stilling basins and constructed wetlands shall be required for large-scale or substantial developments. These include land and community title subdivisions. Design of these devices shall be in accordance with the **Managing Urban Stormwater, Soils and Construction Manual** issued by Landcom and the NSW Department of Environment and Climate Change, March 2004 and June 2008.

- SC11 Silt Traps - Silt traps are to be installed in all stormwater pits to contain silt and debris. Silt traps shall be installed at the bottom of pits at a depth of 200mm to capture silts and fines. Weep holes shall be drilled into the base of the pit to ensure that it does not permanently hold water and create a breeding ground for insects. Where the pit is located over impervious material, a 100mm layer of gravel bedding with subsoil drains will also need to be laid.

Soil and Water Management

- SC12 Soil and Water Management Plan - A soil and water management plan is required for all development types, except for minor developments, for example extension to an existing dwelling of less than 50sqm.
- Minimum guidelines shall be in accordance with the **Managing Urban Stormwater, Soils and Construction Manual** by Landcom and the NSW Department of Environment and Climate Change, March 2004 and June 2008.
- SC13 Installation and Maintenance - All sediment control devices are to be installed prior to any commencement of clearing and earthworks on the site. Ongoing maintenance of these devices during construction will be required.
- SC14 Maintenance Schedule - For large scale developments, residential flat buildings and mixed used developments, a maintenance schedule is to be kept on site to ensure that the devices are cleaned on a regular basis.

Rainwater Re-use

General

- RR1 Application - Notwithstanding BASIX water conservation targets, the collection of rainwater for non-potable use is encouraged. Non-potable water usage shall include watering of the garden, irrigation, washing machine and toilet flushing.

Minimum Requirements

- RR2 BASIX Certificate Requirements - Rainwater re-use shall be provided in accordance with BASIX Certificate requirements.
- BASIX is a web-based planning tool designed and developed by Department of Planning in association with other government agencies and utilities to assess the water and energy efficiency of new residential developments.
- The NSW Government introduced BASIX into the development approval system on 1 July 2004.
- RR3 Rainwater Re-use - Part of BASIX requires the installation of rainwater re-use facilities. The volume of storage required depends on the size of the dwelling, the number of amenities and other site factors. Further details can be obtained at www.basix.nsw.gov.au.
- RR4 Submission of a Development Application - Where it is proposed to install a rainwater tank only, a Development Application (DA) would need to be lodged if the volume of the proposed rainwater tank exceeds 10,000 L. No DA would be required if the proposed rainwater re-use volume is less than 10,000 L or if the rainwater harvesting proposal forms part of a DA for Development Types 2, 3, 4, 4a, 4b, 5, 6 and 7.
- RR5 Dual Usage - Dual usage for both OSD and re-use in the same tank is permitted. However it must be noted that stormwater collected for re-use must not include runoff collected from on-ground areas, balconies and sub-floor areas (sub-surface water). Only stormwater collected from roof areas can be stored for re-use.

- RR6 Permitted Use - Rainwater collected for re-use is classified as non-potable and is only permitted for use in the flushing of toilets, laundry clothes washing, irrigation systems including watering the garden and must not be used for human consumption in accordance with Sydney Water guidelines.
- RR7 When designing dual-purpose tanks, it is essential to consider the collection of rainwater from the site. It is important to note that OSD control typically applied to the entire site area. However, for the re-use component, stormwater can only be collected from the roofs of dwelling/s. A dual-purpose OSD/Rainwater reuse system that solely collects rainwater from the roof may lead to a situation where a significant portion of the surface stormwater runoff from the site bypasses the OSD system. In this regard, the design must carefully account for the majority of the site's runoff, combined stormwater management system is recommended, particularly from roofed and surface areas. This consideration ensures that the runoff can be effectively controlled through the OSD system to meet the overall permissible site discharge requirements.

Stormwater Drainage Design

General

SW1 Application - This Section provides the minimum design guidelines for both private and public stormwater drainage systems.

Stormwater drainage systems referred to herewith are the systems which are designed to convey stormwater runoff. This includes property drainage, street drainage (both piped and surface flow paths), and trunk drainage (larger conduits, open channels) and receiving waters (rivers, creeks, groundwater storage, sea and ocean).

SW2 Purpose - Stormwater drainage systems are to be designed to collect and convey stormwater runoff from a site to a receiving waterway with minimal nuisance, danger to life or damage to properties.

Minimum Requirements

SW3 Australian Rainfall and Runoff Handbook - Stormwater drainage systems shall be designed in accordance with the Australian Rainfall and Runoff Handbook utilising the "major" and "minor" system design criteria.

SW4 Gravity Drainage - Stormwater runoff shall be conveyed to follow the natural fall of the land, by gravity means, wherever possible. The design of "Charged Systems" is not covered in this Section. Charged systems are not acceptable for public drainage systems.

SW5 Re-directing Stormwater - Re-directing stormwater runoff from one catchment (or sub-catchment) to another catchment (or sub-catchment) is generally prohibited.

SW6 Public Drainage systems - Public stormwater drainage systems must be designed as gravity systems. For private stormwater drainage systems, an alternative to gravity drainage will only be considered where easement acquisition is unsuccessful and on-site disposal such as OSA is not possible.

SW7 Minimising the Quantity and Improving the Quality of Stormwater Runoff - Stormwater quantity and quality (pollutants) shall be minimised by designing the system with the following criteria:

- Have adequate sub-surface drainage to provide protection to structures, and prevent long term water ponding
- Have adequate inlets to collect and convey surface stormwater runoff to prevent water from entering buildings or damage structures, minimise nuisance and danger to persons and vehicular traffic, prevent long term surface water ponding, prevent erosion, and protect adjoining and downstream properties from any adverse impacts as a result of stormwater runoff from proposed developments
- Include a system of overland flowpaths, where possible, to provide fail-safe protection to buildings, structures, adjoining and downstream properties in the event of pipe blockage or storm events that generate greater runoff than the capacity of the piped drainage system
- Include sediment and silt traps and trash screens strategically located in catch drains and inlet pits to capture pollutants
- This can be achieved by:
 - The construction of surface flow routes to convey floodwaters away from private and public properties and, in flood prone land, the velocity and depth of flows are controlled to an acceptable level.
 - The provision of surface flow routes and piped drainage systems to direct/control frequent runoff, so that convenience and safety for pedestrians and vehicle traffic can be provided.
 - The provision of both piped drainage and surface flowpaths for new developments, re-developments and new subdivisions.
 - The installation of water quality control devices such as gross pollutant traps, stilling basins, baskets and the like to collect pollutants present in the stormwater runoff.

Minor and Major System Design

SW8 "Dual Drainage" System - The method of stormwater drainage design used to size pipe networks and trunk drainage systems to convey stormwater runoff to the receiving water is known as the "dual drainage" system approach or minor/ major design concept, as described in the Australia Rainfall and Runoff Handbook. This method has been adopted by Council and a summary of the design criteria are set out below.

Minor System Design

SW9 Definition - The gutter and pipe network capable of conveying stormwater runoff during minor storm events is known as the "minor system".

SW10 Widths of Flows - When designing the minor system, adequate inlet pits are to be installed such that the maximum gutter flow widths shall not exceed the lesser of 2.5 metres or one-quarter of the road carriageway width.

Wider flow widths may be accepted on roads whose cross-sectional gradients are less than 1.0%. This is subject to Council approval.

For private stormwater drainage systems, the minor system must be capable of capturing or containing the runoff produced from the average rainfall recurrence interval as stipulated in **SW21 Average Recurrence Interval for Public Drainage Systems**.

SW11 Minimum Conduit Size - The minimum conduit sizes for the minor system design shall be as follows:

- Public system - pipes, 375mm-dia
- Public system - box culverts, 600mm wide x 450mm high
- Private system - pipes, 90mm-dia for roof runoff only
- Private system - pipes, 100mm-dia for surface runoff within the property only
- Roads - pipes, 375mm-dia
- Roads - conduits, equivalency to 375mm-dia

SW12 Minimum Pipe Gradient - The minimum pipe gradient for all types of pipe material shall be 1%.

SW13 Anchor Blocks - Pipes which are to be laid at a slope greater than 20% will require anchor blocks at the top and bottom of the section and at intervals of not more than 3.0 metres in between. Bulkheads shall be provided on steep gradients where soil instability is likely. To retain backfill material, bulkheads shall be installed at intervals of not more than 5.0 metres.

SW14 Pipe Material and Jointing Type - Type of pipe material and jointing to be used for the minor system design shall be as follows:

- Public system - steel reinforced concrete with rubber ring jointed. (Approval required for other material, subject to Council's discretion)
- Private system - up to 300mm-dia uPVC, solvent welded joints; 300mm-dia and larger reinforced or fibre reinforced concrete*, rubber ring jointed

SW15 Depth of Cover - Depth of cover to the conduit obvert shall be as follows:

- Public system not subject to vehicle loads - 450mm
- Public system subject to vehicle loads - 600mm
- Private system not subject to vehicle loads - 300mm
- Private system subject to vehicle loads - 600mm

SW16 Pipe Class - The appropriate class of pipe to be used shall depend on the minimum cover provided and the loading onto the pipe in accordance with **AS3725 - 2007, loads on buried concrete pipes**. The minimum pipe class for Council's public system is Class 4.

SW17 Design Velocity of Flows - Design velocity of flows shall be as follows:

- Conduits - 0.6 m/s minimum, 6.0m/s maximum
- Surface flow - 2.0m/s maximum

Where these values are exceeded, appropriate erosion control and scour protection measures are to be provided at the outlet.

SW18 Maximum Depth of Flows - Maximum permitted depth of surface flow shall be:

- Road carriageway, driveways, footpaths and carparks - 150mm
- Landscaped areas - 300mm

SW19 Annual Exceedance Probability (AEP) for Public Systems - The annual exceedance probability to be adopted for the design of the public drainage system is tabulated below:

Location	AEP (%)
Local Roads	10
Collector roads	10
Sub-arterial roads	10
Arterial roads	5
State roads	2
Access to emergency facilities	1
At a depression (low point) in road	1
Public system passing through private property	1

SW20 Annual Exceedance Probability (AEP) for Private Systems - The annual exceedance probability to be adopted for private drainage systems (minor system design) is tabulated below:

Location	AEP (%)
Residential - low density	10
Residential - medium to high density	5
Commercial	2
Industrial	2
Hospitals and Emergency Facilities	1

*** Important Note:** The underground drainage system (includes pipes, conduits and pits) is to be designed for the 1%AEP storm where major system flows (e.g. along a drainage easement) are likely to surcharge across private property or cause localised flooding. A surcharge path must also be provided to safely convey surface stormwater across private property within easements. The minimum design AEP for surcharge paths shall be the 1%AEP storm event.

Major System Design

SW21 Definition - The "major system" comprises the drainage route, which conveys the runoff for the major storm events. This may be a series of trunk drainage systems and overland surface routes including open channels, creeks, and river systems.

SW22 Provision of Overland Flow Routes - Overland flow routes are to be provided at the following locations:

- Within the road carriageway excluding footpaths and the footway reserve. Flows across footpaths will only be permitted where this will not cause flooding to property or create danger to pedestrians and is subject to Council approval.
- Within drainage easements. Where it is not practical to provide an overland flow route over the easement, the piped drainage system shall be sized to accept the runoff for the major storm event i.e. the 1%AEP.
- Within a known and designated floodway
- Within creeks and river systems.

SW23 Minimum Freeboard Requirement for Flood Prone Land - Freeboard can vary according to parameters such as the size of the upstream catchment, the local terrain, the difference between the 1% AEP (Annual Exceedance Probability) level and the PMF (Probable Maximum Flood) level and the proposed use.

In some cases, the determination of an appropriate freeboard will be best done on the merits of the individual case through the Development Application process. However, the minimum freeboard requirement unless overridden by a Flood Study, Draft Flood Study, Floodplain Risk Management Study, Floodplain Risk Management Plan or the DCP (Development Control Plan) shall be as follows:

Freeboard requirements above 1% AEP water surface level				
Finished Floor Level (B)	Adopted Flood planning area	Overland flow path identified by Council as "Minor**"	Overland flow paths other than ones identified by Council as "Minor"	Mainstream flooding
Residential – Habitable rooms	As per the adopted Plan	300mm	500mm	500mm
Residential – Non-habitable rooms	As per the adopted Plan	300mm	300mm	500mm
Commercial or Industrial – All internal areas	As per the adopted Plan	300mm	500mm	500mm
Carport open on 3 or 4 sides (At Ground Level)	As per the adopted Plan	150mm*	150mm	300mm
Entrance to Basement	As per the adopted Plan	150mm to 300mm*	500mm and/or the PMF Level, whichever is greater	500mm and/or the PMF Level, whichever is greater
Critical Infrastructure	As per the adopted Plan	300mm	500mm and/or the PMF Level, whichever is greater	500mm and/or the PMF Level, whichever is greater

NOTE

* At Council's discretion, may be reduced

** Average depth of flow path affecting the subject property is less than 100mm

SW24 Design Velocities and Depths of Flow - Design velocities and depths of surface flows shall be in accordance with Figures 1 and 2 of the ***New South Wales Flood Risk Management Manual FB03 – Flood Hazard***, with hazard category classed as "H1".

SW25 Annual Exceedance Probability (AEP) - The AEP to be adopted for the major system design is tabulated below:

Location	AEP (%)
Private	1
Public	1

Catchment Area

SW26 Definition - The catchment area is defined by the limits from where surface runoff will make its way, either by man-made or natural paths, to the point of exit.

Catchment Runoff

SW27 Catchment Runoff - Catchment runoff shall be determined using a suitable hydrological method depending on the level of accuracy required and the extent and shape of the catchment. Limitations of each method are given in the Australian Rainfall & Runoff Handbook.

SW28 Determination of Catchment Runoff - Catchment runoff is to be calculated using the following methods:

Method	Application
Rational Method	For catchments areas < 1200 sqm
Hydrological Method	For catchment areas > 1200 sqm

SW29 Rational Method Approach - The Rational Method for the estimation of peak flows will be acceptable for small catchments. This method is best suited to catchments with uniform slope and roughness characteristics and where the level of accuracy is not critical. This would include site discharge calculations and roof runoff.

SW30 Hydrological Computer Model Approach - The use of hydrological computer models is most suitable for medium to large catchments and where a reasonable level of accuracy is required.

Acceptable computer models include ILSAX, DRAINS. Any electronic models utilised in the report and plans for the development application or seeking approval for the activity (S68 – Local Government Act 1993) must be submitted to Council.

SW31 Impervious Areas - The impervious area percentages to be adopted for determining runoff are tabulated as follows:

Public System (trunk drainage)	
Residential Areas	80
Commercial	100
Road Reserves	80

Private System (inter-allotment drainage)	
Road Reserves	80
Residential Single	80
Residential Medium Density	90
Residential High Density	100
Industrial	100
Commercial	100

SW32 Roughness coefficients - Roughness Coefficients are to be used to calculate free surface flow.

For the purpose of determining stormwater runoff into open channels and free surface hydraulics, Manning's roughness coefficients shall be used. Typical values are given in appendix and the Australian Rainfall and Runoff Handbook.

For sections with composite roughness values, Horton's Equation may be used to convert to an equivalent roughness value for simplicity in calculations.

Horton's Equation is given as:

$$n = \left\{ \sum (P_i n_i^{3/2}) / \sum P_i \right\}^{2/3}$$

Where n_i = is the Manning's Roughness Coefficient for section i and

P_i = is the wetted perimeter or length of the section with a roughness value n_i

Hydraulic Grade Line

SW33 Analysis - Hydraulic grade line calculations are to be performed in accordance with the ***Australian Rainfall and Runoff***, and shall be undertaken by a qualified person with experience in hydrology and hydraulic design.

SW34 Full Hydraulic Calculations - Full hydraulic calculations must be submitted for all public and major piped systems (375mm-dia and larger), drainage lines through easements and flood assessment to ascertain flood levels or where Council deems it necessary to determine the feasibility of the proposal.

SW35 Reduction of Losses - Drainage lines shall be designed with minimal bends to avoid bend losses. Where this is unavoidable, junction pits shall be provided at the location of bends or changes in direction.

SW36 Avoidance of Chokes - Pipes will not be permitted to be laid such that a larger pipe joins into a smaller pipe downstream, to avoid potential chokes in the system. However, this may be unavoidable when the new line is connected into an existing system. In this circumstance, the starting hydraulic control shall be adopted at the ground level at the point of connection.

SW37 Water Surface Profiles – For determination of flood profiles and surface drainage systems (e.g. channels, open drains and the like), hydraulic calculations are to include determination of water surface profiles and backwater effect using suitable computer models such as DRAINS, HEC-RAS and TULFLOW. Any electronic models utilised in the report and plans for the development application must be submitted to Council.

SW38 Frictional Losses - Frictional losses in closed conduits of circular shaped cross-section (e.g. pipes), shall be determined using the Darcy-Weisbach Formula. This Formula may be applied to rectangular sections (e.g. box culverts), by converting the product of the area and hydraulic radius to the power of two-thirds ($A.R^{2/3}$) to an equivalent circular section.

SW39 Colebrook-White Roughness Values - Colebrook-White roughness parameters used for pipes are as tabulated:

Conduit Material	K value (mm)
uPVC pipe	0.03
Reinforced concrete pipe	0.06
Fibre reinforced concrete pipe	0.06

SW40 Pit Energy Losses - Pit energy losses and pressure changes at junctions, bends, transition structures, slope junctions, inlet pits, junction pits, drops and outlets must be considered in the hydraulic assessment.

Pressure head coefficients for determining these "head losses" are to be obtained from the following sources:

- Missouri Charts
- Hare Equations
- U S Corp of Engineers mitre bend charts
- AR&R 2019 Handbook or later editions

SW41 Downstream Hydraulic Controls - The following downstream water surface level or controls are to be adopted:

- Where the hydraulic grade line level downstream of the proposed works, including the upstream pit losses at the starting pit is known (corresponding to the design storm recurrence interval as adopted), this level is to be used
- Where the downstream starting point is at a pit and its hydraulic grade line is unknown, a level of 150mm below the surface level of this pit is to be adopted
- Where the outlet is to an open channel, the water surface level is to be determined using Manning's Equation or an appropriate hydraulic model. The water level shall be adopted as the normal depth calculated using an appropriate method, or the top of the outlet pipe, whichever is the greater
- Where the outlet is to an open channel, and downstream flood levels are known, the water surface level to be adopted shall be the 1% AEP flood level

- Where the outlet is affected by tidal or wave action, the resulting tide water level shall be adopted
- Where the outlet is to an existing pipe or conduit, the hydraulic grade line of the conduit shall be determined downstream to a pit where its water level is known (e.g. at a grate in the gutter where the maximum free surface ponding is at the top of kerb)
- Where the outlet is at the invert of the kerb, the water surface level shall be adopted at the top of the kerb

Stormwater Drainage Pits

SW42 Pit Types - Standard Council kerb inlet pits with kerb lintel openings, junction pits and letter box type pits are to be installed for all public drainage systems. Pits for private drainage systems are to be inlet and junction pit types with grates or solid covers.

SW43 Material - Public drainage pits are to be precast concrete, fibre reinforced concrete or insitu poured concrete.

Private drainage pits may be constructed from pre-fabricated high density poly-plastic of appropriate load class as specified by the Manufacturer.

The bases of pits shall be concrete benched to minimise hydraulic losses.

Pits constructed of brick or blockwork will not be acceptable unless they are designed and constructed to withstand structural loading and fully waterproofed to prevent leakage.

SW44 Minimum Dimensions - Minimum internal dimensions of pits are tabulated below:

Pit Type	Dimension (mm)
Kerb Inlet with Lintel and Grate	900 x 900
Inlet Pits in Landscaped Areas	600 x 600
Junction Pits	600 x 600
Private Stormwater and Inlet Pit < 0.60m deep	450 x 450
Private Stormwater and Inlet Pit $\leq 0.60\text{m} \leq 0.90\text{m}$ deep	600 x 600
Private Stormwater and Inlet Pit $\leq 0.90\text{m} \leq 1.20\text{m}$ deep	600 x 900
Private Stormwater and Inlet Pit > 1.20m deep	900 x 900

SW45 Private Stormwater Drainage Pit Location - Private pits are to be located wholly within property boundaries.

SW46 Grates - Grates over public pits are to be galvanised and hinged to frame. Private pits may have grates made from pre-fabricated high density poly-plastic of appropriate load class as specified by Manufacturer.

SW47 Load Class of Grates and Covers - Stormwater drainage pit grates and covers shall be provided to support the required load classification at the following locations:

Location	Load Class Required*
Driveways including access handles	Heavy Duty
Car parks	Heavy Duty
Road Carriageways	Heavy Duty
Footways	Heavy Duty
Reserve, landscaped areas with pedestrian traffic only	Medium Duty
Reserve, landscaped areas with no traffic	Light Duty

* **Important Note:** Light duty is equivalent to 1 tonne wheel load, medium duty is equivalent to 3.5 tonne wheel load and heavy duty is equivalent to 9 tonne wheel load.

All public grates shall be hinged to frame and lockable.

Private grates are to be hinged to frame and lockable where it is designed to surcharge or where upwelling is likely.

SW48 Lifiable Lids - All lids over pits must be liftable for routine inspections and maintenance.

SW49 Junction Pits - Junction pits are not to be designed as pressurised systems.

SW50 Step Irons - Where pits are 1.2 metres or greater in depth, step irons in accordance with **AS1657** are to be provided to one side of the pit wall to allow access for inspections and cleaning.

SW51 Letterbox Type Pits - Letterbox type pits may be used for collection of surface water. However, they are not permitted at or near kerb locations where there is likely pedestrian and/or vehicular traffic. In these locations, pits with butterfly type grates flush with the surrounding ground level are preferred.

SW52 Lintels - Lintels over kerb openings shall be provided at the kerb side with the opening length dependent on its design capture rate, but not less than 1.2 metres and not greater than 4.0 metres.

SW53 Pit Locations - Stormwater drainage inlet pits shall be positioned at the following locations:

- Spaced such that the gutter flow width is limited to 2.5 metres maximum width for the minor system design.
- Located at the upstream side of allotments to minimise runoff flowing across the road.
- Located at sag points and at road depressions.
- Located where access for inspections and maintenance is readily available.
- Provided at changes in direction, grade, conduit level, size, or class of conduit.
- Provided at junctions.
- Spaced at a distance of no greater than 75 metres apart.

SW54 Inlet Capacities – The inlet capacity of pits shall be determined in accordance with the Australian Rainfall and Runoff Handbook.

Inlet capacities are to be restricted with the following blockage factors:

Location	Inlet Type	Capacity Allowed (%)
sag	side entry only	80
sag	grate only	80
sag	combination	100 side 0 grate
sag	letterbox	50
on-grade	side entry only	80
on-grade	grate only	50
on-grade	combination	80

Open Channels

SW55 Preferences – Piped drainage systems are preferred over open channel systems.

Open channels will only be permitted if they form part of the major drainage system and where permitted, shall be designed to have smooth transitions, with adequate access provisions available for inspections, general maintenance and adequate safety measures installed to protect persons and vehicles.

These include perimeter fences, bollards, and grills over outlet pipes. Step irons shall be installed and flow velocities and depths are to be reduced at nominated access points.

SW56 Design – Open channels shall be designed in accordance with the Australian Rainfall and Runoff Handbook and the NSW Government Floodplain Management Manual and must be able to contain the major system runoff. Open channels shall be designed to avoid hydraulic jumps or generate supercritical flow conditions. Side slopes must not exceed 1 in 3, unless fully fenced off.

Wherever possible, low flows shall be contained within a piped system or contained within a concrete lined channel at the invert of the channel.

SW57 Manning's Roughness Coefficient – Manning's roughness coefficients for open channel sections applicable to specific channel types can be obtained from the Australian Rainfall and Runoff Handbook.

Typical values are given below:

Surface Finish	n value
Concrete pipes or box sections	0.012
Concrete trowel finish	0.015
Concrete formed without finishing	0.016
Sprayed concrete, granite	0.018
Bitumen, smooth finish	0.016
Bricks or pavers	0.016
Pitchers or dressed stone in mortar	0.016
Rubble masonry or random stone in mortar	0.028
Rock lining or rip-rap	0.028
Earth, clean	0.027
Corrugated metal	0.027
Earth, weed and gravel	0.022
Rock cut	0.028
Short grass	0.033
Long grass	0.035
Medium to dense brush	0.150

Building Adjacent to Stormwater Drainage Systems

SW58 Location of Structural Supports - Where structural supports such as foundations, piers, and footings are to be located adjacent to the drainage system, they are to be located outside of the drainage easement and must not load bear onto the underlying drainage structure. In general, where a drainage structure is parallel or adjacent to foundations, piers or footings, the base of the footing shall be located outside the influence zone created by an angle of 45 degrees to the horizontal extended from the invert of the drainage structure.

SW59 Where a public infrastructure drainage line or a drainage easement is identified within a development site, the location of the public line and/or drainage easement must be accurately located on the development application and construction certificate drawings.

Stormwater Connections

SW60 Stormwater connections from private land – any proposed or existing stormwater connection from private land to the Council's drainage system is a private stormwater connection. It is the responsibility and liability of the private landowners to maintain the private pipeline that crosses the footpath area is in good condition at no cost to Council.

SW61 Under Buildings - shall be carried out in accordance with the latest **AS/NZS3500.3**.

SW62 Above ground pipe work - Shall be carried out in accordance with the latest **AS/NZS3500.3** and this Document. The stormwater discharge point's location must be confirmed by Council.

Non-return valves are not permitted to be installed in Council maintained (public) systems.

SW63 Connection Detail - If the ratio of the private pipe size to the Council (public) pipe size is more than one third, a standard stormwater gully pit at the connection point, will be required.

Avoid Conflict with Utility Services

SW64 General - Care shall be taken to ensure that the proposed stormwater drainage system will not conflict with utility services. In this regard, all utility services shall be located prior to final drainage system design. Stormwater drainage conduits crossing over or under sewer lines must be laid in accordance with Sydney Water requirements. This may require support trenching and concrete encasement of sections that traverse the utility.

SW65 Location of Drainage Lines in Road Reserve - Public drainage lines laid in the road reserve, shall be located under the kerb line or within the road carriageway, to avoid conflict with the utility services in the footway.

For private drainage lines which must cross the footway reserve, they shall be laid across the footway perpendicular to or at a maximum angle of 45 degrees to the kerb face to minimise conflict with services. The stormwater connection point must be wholly within the subject site frontage.

Easements

SW66 Requirement - Where the site grades to the rear, the creation of formal drainage easements will be necessary. Evidence in the form of a Legal Agreement between affected parties or copies of titles showing the creation or intention to create easements must be provided to Council in support of the Development Application. This is essential for DA approval. Where easements cannot be negotiated, signed documents by all affected parties or acceptable documentation provided as evidence shall be submitted to support the case for alternative solutions.

Easements shall be required over constructed public drainage systems within private properties, to **ensure** that Council has full rights of access to such drainage systems for the purpose of inspection, maintenance or upgrade.

Stormwater drainage easements shall be required over all private inter-allotment drainage lines.

Stormwater drainage easements will not be required within or over natural drainage systems such as creeks and watercourses.

Council does not favour or encourage the piping, construction within or over, or interference with natural drainage systems. Any proposal to carry out such works will be subject to an assessment pursuant to other relevant Council policies and joint approval from other relevant authorities.

SW67 Easement Acquisition - Council will take the opportunity to acquire drainage easements over existing constructed public drainage systems within private property whenever a development occurs by Condition of Consent.

When a developer, property owner or Council proposes to relocate or reconstruct a public drainage system within the site, a drainage easement in Council's favour must be created to suit the relocated or reconstructed drainage system at the cost to the developer.

A subdivision **will not** be approved where a formal drainage easement cannot be provided to drain a newly created allotment in the same direction as the natural fall of the land.

SW68 Prohibition of Building over Easements - Any construction of buildings or other permanent structures over a public drainage easement is not permitted.

Easements allow Council reasonable access for the purpose of construction, maintenance and upgrade of the stormwater drainage system.

Encroachments impede or re-direct overland flows away from the easement and may load bear onto the underlying drainage structure.

Council may require the relocation of an existing pipe to avoid encroachment. Where encroachment is necessary to achieve reasonable development, then the conduit and easement may be relocated subject to Council approval.

SW69 Load bearing Structures adjacent to Drainage Easements - Any structural support such as footings and piers will only be permitted to be located adjacent to an easement if they do not load bear onto the underlying drainage structure, and that the built structure will not be undermined by any future maintenance work necessary within the easement.

SW70 Planting of Significant Trees - The planting of trees or large shrubs, particularly those with extensive root systems will not be permitted in or within proximity of drainage easements.

SW71 Construction over Public Drainage Easements - Construction over a public drainage easement is generally prohibited. Public drainage easements must be free of all encroachments with a minimum vertical clearance from the surface/ground level over the drainage structure to a height of 6.0 metres above.

Paved surfaces over the public drainage easement are permitted provided that construction joints along each longitudinal edge of the easement are installed to facilitate access to the drainage structure.

SW72 Public Easement Widths - Public easement width shall be a minimum width equal to the external width of the conduit plus 1.0 metres on each side, rounded up to the nearest 0.1 metres.

In all cases, authority to modify or extinguish the easement shall be vested in the City of Canada Bay Council.

Inter-allotment Drainage

SW73 Application - Inter-allotment drainage shall be provided for property/s that does not drain directly to its street frontage by gravity means, or directly by gravity means to an existing stormwater drainage system (public or legally private) or by gravity means to a natural watercourse.

Easements shall be created over all inter-allotment drainage systems on private property/s in favour of all upstream properties to be benefited by it.

SW74 Inter-allotment Easement Widths – Inter-allotment easement widths (private) shall be a minimum width of 0.9m. However these widths may be varied where there are site constraints.

SW75 Sizing of Inter-allotment Drainage Systems - Inter-allotment conduits shall be sized on the total site area of the benefitted allotments with a minimum impervious area as given in Section **Impervious Areas**.

SW76 Material - Inter-allotment drainage pits and pipes shall be of reinforced concrete, fibre reinforced concrete or uPVC material. All pipes/ conduits are to be rubber ring jointed or solvent welded as specified by the manufacturer and in accordance with **AS4058**, **AS4139** and **AS1254** respectively.

Subsurface Water

SW77 Definition - Subsurface or groundwater is water held in the soil or in pores and crevices in rock and is generally present behind basement walls or subfloor areas which are below the natural ground level.

SW78 Minimum Information to be Provided – Where a proposal is for the installation of a basement or below ground area or the development will require cutting into the existing ground level, a Geotechnical Report must be submitted to Council for assessment. This Report shall detail an investigation of subsurface conditions including groundwater levels, the likelihood or indication of a high water table or seepage water, the soil type encountered and the soil infiltration rate.

SW79 Disposal – Generally subsurface water is collected by sub-soil drainage pipes and collected in a holding tank which is either pumped or gravity fed into an approved disposal point. Subsurface water is not permitted to be discharged to the kerb either directly or indirectly except in a controlled manner specifically approved by Council under **S138.1(d) of the Roads Act 1993** (refer **SW80** below). It must be drained directly into an underground public drainage system or other disposal point approved by Council.

SW80 Direct Connection into an Underground Public Drainage System – This is the preferred method of subsurface water disposal. The subsurface water is conveyed across the public footway by gravity means and drained into the nearest public underground piped drainage system.

A junction pit is to be installed within the property boundary with a non-return valve on the upstream pipe to prevent water back flowing into the site.

If there is no existing underground drainage system in the vicinity of the source, a new underground (piped) drainage system shall be installed and connected into the nearest downstream underground drainage system. The new drainage system other than the line that crosses the footway shall be designed to a public drainage system standard and shall become part of the public drainage system.

SW81 Waterproofing Subfloor Areas – Basement walls and subfloor areas may be waterproofed to minimise the collection of seepage water, in particular where Type 2 acid sulphate soils or high water tables are encountered.

Water proofing has its limitations and subsoil water ingress may still exist as there would be high pore pressures exerted on the walls and floors of the “tanked” structure. Pore pressures and floatation forces mean these types of structures require specialized design and certification by a Qualified Structural Engineer.

SW82 High Water Table – In the case of a high water table, the draw down effect of the water table when subsurface water is collected and disposed would necessitate a statement from a Geotechnical Engineer that nearby structures will not be affected by the development. It may be necessary to obtain licence for the drawing down of groundwater from the Department of Primary Industries.

- SW83 Controlled Disposal – If all alternative acceptable solutions have been exhausted, Council will consider separate specific approval of controlled release to the kerb under **Section 138.1(d) of the Roads Act**. In this circumstance subsurface water may be disposed to the kerb and gutter in a controlled rate, period and duration. This will necessitate the following:
- Provision of a sump or holding tank which can store collected seepage water for a minimum duration of 24 hours.
 - The soil infiltration rate shall be determined by geotechnical investigation but shall not be less than 0.001 L/s per m².
 - The holding tank shall be sized for the collected seepage water based on the area of exposed wall to the soil. For example, for typical double garage and an infiltration rate of 0.001 L/s per sqm and a wall height of 2.4m with subsoil drainage install around its perimeter say 25m long, the volume of the holding tank required will be $2.4 \times 25 \times (0.001/1000) \times 24 \times 60 \times 60 = 5.2$ cum or 5200 L required. Note this is in addition to the volume required for collected stormwater from weather exposed areas.
 - Dual pumps to be installed and shall be designed to operate to discharge a maximum of 5L/s and restricted to pumping between 11pm and 3am only.

Stormwater Pollution and Erosion Control

Soil and Water Management

- SPE1 Soil and Water Management Plan - A soil and water management plan shall be submitted and must be approved by Council prior to the commencement of any construction or demolition activity.
- SPE2 Guidelines - Guidelines shall be in accordance with the ***Managing Urban Stormwater, Soils and Construction Manual*** issued by Landcom and the NSW Department of Environment and Climate Change, March 2004 and June 2008.
- SPE3 Sediment Control Devices - All sediment control devices are to be installed prior to any commencement of clearing and earthworks on the site. Ongoing maintenance of these devices during construction will be required.
- A maintenance schedule is to be provided with the Soil and Water Management Plan for large scale developments and works within public land.
- SPE4 Soil Erosion Control - Soil erosion control is required to protect adjoining properties, bushland, roadways and receiving waters from degradation due to silt laden stormwater runoff as a result of development and/or concentration of runoff. Soil erosion control shall be provided as follows:
- Appropriate scour protection installed at the outlet to stormwater conduits, and
 - Installation of pollution control devices at the source, on-line, off-line or at the end of the line to control sediment laden overland stormwater flows.
- SPE5 Scour Protection Devices - Scour protection devices shall include embankment stabilisation e.g. rock walls, concrete aprons, gabions, turf, jute mesh, energy dissipating units, or other more appropriate erosion control devices approved by Council.
- Please note that some types of scour protection devices may be inappropriate along certain creek locations.

- SPE6 Silt and Sediment Control - Silt and Sediment Control devices are required as part of a development to remove pollutants during the "first flush".

These units shall be installed on line or within the site and may include proprietary items such as gross pollutant trap, hydrodynamic separation, filters, basket or silt and grease arrestors approved by Council. Installation of these devices shall be in accordance with the manufacturer's specification.

Other sediment control devices such as stilling basins and constructed wetlands shall be required for large-scale developments and include land and community title subdivisions.

Design of these devices shall be in accordance with the ***Managing Urban Stormwater, Soils and Construction Manual*** issued by Landcom and the NSW Department of Environment and Climate Change, March 2004 and June 2008.

Integrated Developments

- SPE7 Part 3A Permit for Development near Water Bodies - Any development within 40m of a water body such as a stream, creek, lagoon, or river may require a Part 3a Permit under the Department of Lands Guidelines and The Rivers and Foreshores Improvement Act 1948.

Water Sensitive Urban Design

Objectives

WSUD1 Principles of Water Sensitive Urban Design
- All developments are encouraged to implement the principles of Water Sensitive Urban Design (WSUD) in order to minimise the impact of the development on the water cycle and achieve more sustainable forms of urban development.

WSUD2 Aim of WSUD - The aim of WSUD is to integrate stormwater management systems into the landscape component of the site in a manner that provides benefits that incorporate stormwater detention, retention, re-use and water efficiency whilst addressing issues such as nuisance flooding, protection from pollution of the receiving waterways and groundwater and improving visual amenity.

Design Principles

WSUD3 How Design Principles are achieved - In addition to those control systems already identified in **Stormwater Management** such as On-site Stormwater Detention, infiltration or absorption systems and rainwater harvesting, the aim of WSUD is to combine these options together with the proposed landscaping to achieve integration in accordance with **Clause SC4 (Pollutant Load to be Retained)** rather than relying on 'end of pipe' devices prior to discharge. These are achieved by:

- Operating practices and technologies which would prevent contamination of stormwater.
- Locate the development such as to minimise disturbance of the natural drainage system.
- Minimise impervious surfaces and encouraging soft landscaping to promote infiltration and reduce stormwater runoff.
- WSUD elements are to be located and configured such that impervious areas to be treated can be maximised.

WSUD4 Water Sensitive Urban Design Measures - Where WSUD principles are to be applied, the following table lists measures, which can be used to achieve water quality and water quantity targets. Note their uses may be complementary to other 'hard' engineering solutions.

WSUD Measure	Comments
Constructed Wetlands	To be considered for medium to large scale subdivisions
Sedimentation Basins	To be considered for medium to large scale subdivisions
Vegetated Filter Strips	Can be incorporated into new developments
Sand Filters	Used in combination with absorption system design
Bio-retention Systems	Used in combination with absorption system design or for water quality improvement targets in lieu of straight discharge into the waterways. Alternative Gross Pollutant Traps may be considered (mechanical system) for screening silts, sands and debris, may be considered
Permeable Pavers	As a complementary system to absorption trench to minimise on-ground impervious areas
Infiltration Trenches	Absorption trench design as described above
Rainwater Harvesting	Re-use is encouraged.

WSUD5 Water Sensitive Urban Design Modelling – Modelling of WSUD using a suitable program such as MUSIC (Model for Urban Stormwater Improvement Conceptualisation) will be acceptable to Council.

WSUD6 Monitoring Water Quality - Water quality treatment systems or pollution control devices are to be installed and monitored regularly to ensure that they achieve their treatment objectives and that their performance meets the above criteria. If they fail to meet these targets, or if so required by the Appropriate Regulatory Authority (ARA), it shall be modified to achieve them and the system upgraded.

WSUD7 Ease of Maintenance - Water quality treatment systems or pollution control devices are to be designed to ensure ease of maintenance. The location of the WSUD measure shall locate where easy unimpeded access is possible for routine inspections and maintenance requirements.

WSUD8 Maintenance Schedule – where the development site requires to provide WSUD strategy and measures as per WSUD4 and/or 'hard' engineering solutions such as gross pollutant treatment devices, filter units etc. A maintenance schedule shall be prepared by the qualified suitable engineer demonstrating a simple set out maintenance actions and the frequency of such actions including a sketch plan to ensure the constructed devices can be maintained by the future property owners and occupiers.

WSUD9 Positive Covenant and Restriction on the Use of Land – The constructed stormwater quality improvement device (SQID) shall require the creation of a Positive Covenant and Restriction on the Use of Land affixed to the title requiring the owner of the property to maintain the SQID system including inspect, removal of the pollutant, maintain and replace the treatment devices in order to maintain the reduction target in accordance with the approved documents. Restriction on the Use of land to ensure the treatment devices will operate/perform as per approved documents and no alter or addition of the treatment devices without written consent from authority.

Document Submission

General

DS1 Minimum Requirements - At the lodgement of a Development Application, minimum submission of documentation is required to support the proposed development. Usually conceptual plans would suffice unless the proposed development would impact on public land or Council considers that there may be site constraints which may result in non-compliance to standards.

DS2 Stormwater Drainage Conceptual Plan - a stormwater drainage concept plan (SDCP) must be submitted with the Development Application.

DS3 Detailed Design Drawings - Detailed design drawings and supporting calculations are generally required at Construction Certificate submission and not required at DA stage except in the following circumstances:

- The proposed works include activities on Council property. A separate approval under Section 138 of the Roads Act may be necessary. It is important the Applicant contact Council in this case to ensure that approval can be granted.

Important Note: Works in the road reserve can only be approved under Section 138 of the Roads Act. Approval under Part 4 of the Environmental Planning and Assessment Act does not grant automatic approval for works on public land.

- The proposed development is a new dwelling, residential flat building, mixed development, commercial or industrial development and the land falls to the rear and does not benefit from a formal drainage easement.
- A Stormwater Assessment Report is required due to likely flood affectation to ascertain design floor levels or if there is a proposed subfloor level which may be likely to be affected by stormwater inundation or disposal of seepage water is difficult.

Where detailed design drawings are required, they must be submitted and approved by Council prior to the issue of the Construction Certificate.

Development Application

DS4 Submission of Conceptual Drawings - Engineering conceptual drawings shall be submitted at the lodgement of the Development Application containing the following information:

- A3 or A1 size drawing sheets at an appropriate scale of 1:100, 1:200 or 1:500.
- Showing the layout of the proposed drainage system and structures including the location of all downpipes, kerbs, channels, open drains, pits, pipes, retaining walls etc.
- Showing the nominal size of all stormwater conduits, grades, and pit dimensions.
- Showing conduit gradients.
- Showing the finished surface levels of any open channels, drains, or swales.
- Showing the location of all buildings, driveways, retaining walls, and other impervious and pervious surfaces.
- Showing the finished surface levels of paved areas, unpaved areas, building floors and garages.
- Showing typical cross sectional details of any open channels, drains, or swales.
- Where drainage easements are required, the location of the proposed easement for stormwater drainage and legal agreements attached.
- Showing details of the proposed On-site Stormwater Detention (OSD) or On-site Stormwater Absorption (OSA) System, Stormwater Quality improvement Device (SQID) or Mechanical Pump-out, if required.
- Include basic supporting computation or information, including any electronic hydrological and/or hydraulic modelling utilised in the submitted engineering report or plans.
- Showing the location of any utility services, structures, trees, etc., which might affect the proposed development.

Important Note: It is the responsibility of the applicant to submit sufficient details of all relevant services, which may conflict with the proposed design. The exact locations of any crossings or connections are to be shown.

Construction Certificate

DS5 Submission of Construction Certificate

Drawings - Detailed construction drawings are to be submitted as required, either to Council or to a nominated Principal Certifier (PC) and must include the following information:

- Drawings to be issued on standard A3 or A1 size drawing sheets at an appropriate scale of 1:100, 1:200 or 1:500.
- Drawings must clearly show the layout of the proposed engineering works, including the location of all retaining walls, downpipes, pits and pipes and labelled with their pipe sizes, gradients, existing and finished surface and invert levels, dimensions of all OSD, OSA, and pump-out systems, including the proposed storage volumes, surcharge/overflow paths, permissible site discharge, and other relevant information sufficient for construction.
- All supporting calculations are to be supplied on computer disc (where applicable) with relevant hydrologic and hydraulic information.

DS6 Minimum Information - the minimum information to be provided on the drawings are as follows:

- Total site area in m².
- Total impervious area (roof and paved) in m².
- Area draining into the proposed stormwater (OSD or OSA) facility.
- Dimensions (mm), volume (cum), and discharge rate from the OSD or OSA system.
- Maximum water depth (mm) from centreline of outlet to top water level.
- Maximum depth of ponding for above ground OSD systems.
- Type and size of orifice (mm-dia), outlet pipe (mm-dia) and PSD (L/s).
- Details of the OSD control device(s) used including size and shape, outlet pipe diameter and invert level.
- For underground systems, at least one (1) detailed section through the OSD or OSA facility sufficient for construction.
- For above ground systems, at least two (2) detailed sections through the OSD facility, which shall include the maximum water level,

gradients, and overflow weir, sufficient for construction.

- Existing and proposed levels and details of adjoining structures and buildings shall be shown on the sections through the OSD or OSA facility.
- Overland flow path and PSD from the site.
- Plan showing the location of OSA or OSD facilities including dimensions, pervious (landscape) and paved (existing and proposed roof and paved) areas, and all existing and proposed surface levels.
- For pump-out systems, the pump type and rate, and holding tank volume.
- Location of any utility services, structures, trees, etc., which may affect the proposed drainage system.
- Full details of all relevant services, which may conflict with the proposed design, including invert levels and size of the service are to be shown.
- Structural details of retaining walls including sub-soil drainage, reinforcement details, dimensions, and concrete strength.
- Where an inter-allotment drainage line is to be laid, a longitudinal section of the proposed pipe from the point of connection to the discharge point is to be provided. This must include pipe sizes, gradients, flowrates, and a hydraulic grade line.

DS7 Design Certification - The construction drawing for all types of development must be designed and certified by a suitably qualified practising Engineer, with membership with the Institution of Engineers, Australia, practising in the relevant competency category (civil, geotechnical, structural) and on the National Engineering Register (NER).

DS8 Drainage Easements - Where drainage easements are required, evidence in the form of a legal agreement between the affected parties or copies of titles showing the created easements shall be submitted with the drawings.

Prior to Occupation (Final) Certificate or Completion of Works

DS9 Minimum Information – Following construction and prior to the occupation of the development, the following document in accordance with the requirement prepared by a suitably qualified person shall be submitted to Council:

- **Works-As-Executed (WAE) drawings:** an engineering survey of the as built works prepared on the certified (or approved) plan issued by the principal certifier or the relevant authority. A copy shall be submitted to Council. This plan is to include constructed levels, dimensions, and volume of the built OSA, OSD facility, the location of all drainage pipes, sizes and levels, etc., and signed by a Registered Surveyor.

Note: WAE plan must be final version. Draft revision or watermark is not accepted.

- **Copies of titles** showing the creation of Positive Covenants and Restriction on the use of land. Standard wording of positive covenants and restriction on the use of land can be found in Section ES4 Appendices.
- **Final compliance certification** of the constructed drainage system by a suitably qualified engineer for all types of development. The certification must certify that the works have been installed in accordance with the relevant Australian standards (e.g., AS3500) and Council's relevant specifications and DCP.
- **Identification Plate:** an identification plate of not less than 110mm wide x 80mm high, is to be fixed near or onto the control structure of the OSD system, this is to advise the registered proprietor of their responsibility to maintain the OSD facility and not to tamper with it in any manner without written consent. This plaque shall read 'This is an On-site Stormwater Detention System. It is an offence to reduce the volume of the system (tank or basin) or to remove the orifice that controls the outflow. The base of the outlet control pit and the debris screen must be cleared of debris and silt on a regular basis. This plate must not be removed.'

Identification plates can be purchased from Council's Customer Services Centre

DS10 Where on-site stormwater detention system, on-site stormwater absorption system, stormwater quality improvement devices or a mechanical pump out system is proposed on site, the following documents must be submitted to Council or Principal Certifier prior to occupation:

- a) A certificate of compliance from a suitably qualified engineer with Institution of Engineers, Australia Corporate Membership and registered on the National Engineering Register (NER) under the appropriate professional Category certifying the intended function of constructed stormwater drainage system including Onsite Stormwater Detention (OSD) system, Onsite Stormwater absorption system, Stormwater Quality Improvement devices, Mechanical Pump out system etc, and
 - b) "Work - As - Executed" drawings of constructed stormwater drainage system prepared by a Registered Surveyor or equivalent.
- Where Council is not the Principal Certifying Authority, two (2) copies of the above documents are to be provided to Council prior to the issue of any Occupation Certificate. These documents are to be retained on Council's Construction Certificate files.

DS11 Works-As-Executed Documentation for the constructed public infrastructure:

Prior to handing over of land or assets to Council, the developer is to provide Works-As-Executed documentation.

The documentation to be provided is to be in the form of:

- a) A certificate of compliance issued by a suitably qualified engineer with National Engineering Register (NER) accreditation for the constructed public infrastructure shall be submitted to Council;
- b) Work-As-Executed (WAE) drawing prepared by and certified by a registered surveyor. The WAE drawing is to be prepared on the certified (or approved) plan issued by the principal certifier or the relevant authority;
- c) A1 size pdf drawings suitable for archiving (vector based, not raster); and
- d) A vector format files suitable for importing into Council's Land Information Systems (Esri Geodatabase or MapInfo) or CAD systems (AutoCAD).
- e) A CCTV (closed circuit television) report and footage verification of the newly constructed public stormwater drainage lines and demonstrating no signs of crack post installation.

Note: The CCTV footage shall include but not limited to the size of pipe, location from point to point (e.g., A to B), material of pipe, date of the works, length of the pipe and pipe gradient. This is to be accompanied with a map or plan of the stormwater drainage pipeline surveyed.

DS12 Detailed Asset Data - Prior to handing over land or assets to Council, the developer is to provide detailed asset data at a component level for importation into Council's Asset Management System.

Information required will depend on the components and the developer should submit a component list to Council for further advice.

The following list is representative of what would be required:

- A vector representation of the component and its location on MGA2020 coordinates linked to:
 - Description.
 - Material details.
 - Design life.
 - Installation date.
 - Acquisition date.
 - Dimensional and quantity information where it is not defined by the graphic object (e.g. The length of a pipe can be represented by the length of a line object on the plan. A data value representing the diameter needs to be attached to that line object. The area of a segment of road can be defined by a polygon. A data value representing the pavement depth needs to be attached to the polygon.)
 - Valuation at installation date.

It is recommended that these requirements be considered at the design documentation stage.

For more complex and proprietary items, for example pumps or lighting systems, the developer shall provide to Council's satisfaction details of Brand, Model, Supplier, Warranty, Installation Guide, User Guide, Safety Instructions, Servicing Record, Maintenance Manual, Keys etc as appropriate.

Section 138 Consent under the Roads Act

DS13 General - Development Approval does not give automatic approval for external works to the site, that is, in public domain areas.

Important Note - Private Certifiers cannot issue Consent under the Roads Act. The Road Authority which can issue Consent under the Roads Act will generally be either Council or the Transport for NSW (TfNSW). In some cases where Council or TfNSW is the Road Authority, the consent of both Council and the TfNSW will be required due to traffic or maintenance impacts.

Council is the Consent Authority for all works within the footway area, regardless of TfNSW approval for road pavement works or activities. This may be in the form of a **Road Opening Permit** (for minor works such as the laying of a private stormwater pipe across the footpath), **Vehicular Crossing Application** (for driveway and footpath construction) or a **Section 138 Application for Major Works** (such as laying a stormwater drainage line under the kerb or road pavement construction). All these activities require a separate consent from Council under Section 138 of the Roads Act 1993.

DS14 External works - Pursuant to Section 138 of the Roads Act 1993, written consent from the Appropriate Road Authority (Council or TfNSW), must be sought for proposed works external to the site. External works include the following:

- Closure of a carriageway on a State Road
- Closure of a carriageway on a Regional Road
- Works which may impact the traffic flow on a State Road or Regional Road
- Works within 100m of a Traffic Facility (e.g. Signalized Traffic Lights)
- Closure of a lane for the purpose of standing a crane, concrete pump or waste bin

DS15 Types of External Works cover by a Section 138 Approval - The types of works requiring Section 138 approval include but not limited to:

- Road works in general
- Stormwater drainage works
- Traffic devices or local area traffic management (LATM) schemes
- Footpath construction
- Water quality control devices on public land
- Driveway construction
- Any related works within an adjacent road reserve (path, verge or carriageway)

DS16 Road Occupancy License - Any works within the road reserve, for a State or Regional classified road, will require a Road Occupancy License to be obtained from the Planned Incidents Unit of the Traffic Management Centre of the TfNSW.

The Application must include details of a Traffic Management Plan.

DS17 Other Consents under Section 138 of the Roads Act - **An Application for Vehicular Crossing Construction and Ancillary Works, Road Opening Permit** or a **Consent Letter** from Council is required for all other types of works within the road reserve.

ES4 Appendices

Terms of Positive Covenant

Existing Allotments

- A1 The Terms of Positive Covenant - Where there is no land subdivision (no Section 88B instrument required) then the following standard wording for the "The Terms of Positive Covenant" is to be used and attached with the standard NSW Department of Lands form 13RPA.

Terms of Positive Covenant

(Show full details of Positive Covenant)

The registered proprietors covenant with the City of Canada Bay Council (Council) that they will maintain and repair the structure and works on the land in accordance with the following terms and conditions:

I. The registered proprietor will:

- i. keep the structure and works clean and free from silt, rubbish and debris
- ii. maintain and repair at the sole expense of the registered proprietors the whole of the structure and works so that it functions in a safe and efficient manner.

II. For the purpose of ensuring observance of the covenant the Council may by its servants or agents at any reasonable time of the day and upon giving to the person against whom the covenant is enforceable not less than two days notice (but at any time without notice in the case of an emergency) enter the land and view the condition of the land and the state of construction maintenance or repair of the structure and works on the land.

III. The registered proprietors shall indemnify the Council and any adjoining land owners against any claims for damages arising from the failure of any component of the on-site stormwater detention (OSD), on-site stormwater retention/absorption (OSA), stormwater quality improvement device (SQID) and/or mechanical pump-out system, or failure to clean, maintain and repair the stormwater management system.

IV. By written notice the Council may require the registered proprietors to attend to any matter and to carry out such work within such time as the Council may require to ensure the proper and efficient performance of the structure and works and to that extent Section 88F(2) (a) of the Act is hereby agreed to be amended accordingly.

V. Pursuant to Section 88F(3) of the Act the authority shall have the following additional powers pursuant to this covenant:

i. In the event that the registered proprietor fails to comply with the terms of any written notice issued by the Council as set out above the Council or its authorised agents may enter the land with all necessary equipment and carry out any work which the Council in its discretion considers reasonable to comply with the said notice referred to in I hereof.

ii. The Council may recover from the registered proprietor in a Court of competent jurisdiction:

(a) Any expense reasonably incurred by it in exercising its powers under sub-paragraph i hereof. Such expense shall include reasonable wages for the Council's own employees engaged in effecting the said work, supervising the said work and administering the said work together with costs, reasonably estimated by the Council, for the use of machinery, tools and equipment in conjunction with the said work.

(b) Legal costs on an indemnity basis for issue of the said notices and recovery of the said costs and expenses together with the costs and expenses of registration of a covenant charge pursuant to Section 88F of the Act or providing any certificate required pursuant to Section 88G of the Act or obtaining any injunction pursuant to Section 88H of the Act.

VI. This covenant shall bind all persons who claim under the registered proprietors as stipulated in Section 88E(5) of the Act.

For the purposes of this covenant:

Structure and Works shall mean the on-site stormwater detention (OSD), on-site stormwater retention/absorption (OSA), stormwater quality improvement device (SQID) and/or mechanical pump-out system constructed on the land as set out in the plan annexed hereto and marked with the letter "A" or alternatively as detailed on the plans approved by the Principal Certifying Authority (INSERT ORGANISATION/COMPANY NAME): {INSERT CONSTRUCTION CERTIFICATE NUMBER AND DATE ISSUE BY PRINCIPAL CERTIFYING AUTHORITY, DRAWING NUMBER, DATE, REVISION NUMBER AND DESIGNER DETAILS} including all gutters, pipes, drains, walls, kerbs, pits, grates, tanks, chambers, basins and surfaces designed to temporarily detain stormwater on the land.

The Act means the Conveyancing Act 1919.

A copy of the construction certificate or complying development certificate is held on Council file "DA number" or "CDC number".

Name of authority having the power to release, vary or modify the 'Positive Covenant' is City of Canada Bay Council.

Land Subdivision

A2 The Terms of Positive Covenant - Where a subdivision has been lodged and a Section 88B instrument created, then the following standard wording for the "The Terms of Positive Covenant" shall be used.

Terms of Positive Covenant referred to in the above-mentioned Plan

The registered proprietors covenant with the City of Canada Bay Council (Council) that they will maintain and repair the structure and works on the land in accordance with the following terms and conditions:

I. The registered proprietor will:

- i. keep the structure and works clean and free from silt, rubbish and debris
- ii. maintain and repair at the sole expense of the registered proprietors the whole of the structure and works so that it functions in a safe and efficient manner.

II. For the purpose of ensuring observance of the covenant the Council may by its servants or agents at any reasonable time of the day and upon giving to the person against whom the covenant is enforceable not less than two days notice (but at any time without notice in the case of an emergency) enter the land and view the condition of the land and the state of construction maintenance or repair of the structure and works on the land.

III. The registered proprietors shall indemnify the Council and any adjoining land owners against any claims for damages arising from the failure of any component of the on-site stormwater detention (OSD), on-site stormwater retention/absorption (OSA), stormwater quality improvement device (SQID) and/or mechanical pump-out system, or failure to clean, maintain and repair the stormwater management system.

IV. By written notice the Council may require the registered proprietors to attend to any matter and to carry out such work within such time as the Council may require to ensure the proper and efficient performance of the structure and works and to that extent Section 88F(2) (a) of the Act is hereby agreed to be amended accordingly.

V. Pursuant to section 88F(3) of the Act the authority shall have the following additional powers pursuant to this covenant:

i. In the event that the registered proprietor fails to comply with the terms of any written notice issued by the Council as set out above the Council or its authorised agents may enter the land with all necessary equipment and carry out any work which the Council in its discretion considers reasonable to comply with the said notice referred to in I hereof.

ii. The Council may recover from the registered proprietor in a Court of competent jurisdiction:

(a) Any expense reasonably incurred by it in exercising its powers under sub-paragraph i hereof. Such expense shall include reasonable wages for the Council's own employees engaged in effecting the said work, supervising the said work and administering the said work together with costs, reasonably estimated by the Council, for the use of machinery, tools and equipment in conjunction with the said work.

(b) Legal costs on an indemnity basis for issue of the said notices and recovery of the said costs and expenses together with the costs and expenses of registration of a covenant charge pursuant to Section 88F of the Act or providing any certificate required pursuant to Section 88G of the Act or obtaining any injunction pursuant to Section 88H of the Act.

VI. This covenant shall bind all persons who claim under the registered proprietors as stipulated in Section 88E(5) of the Act.

For the purposes of this covenant:

Structure and Works shall mean the on-site stormwater detention (OSD), on-site stormwater retention/absorption (OSA), stormwater quality improvement device (SQID) and/or mechanical pump-out system constructed on the land as set out in the plan annexed hereto and marked with the letter "A" or alternatively as detailed on the plans approved by the Principal Certifying Authority (INSERT ORGANISATION/COMPANY NAME): {INSERT CONSTRUCTION CERTIFICATE NUMBER AND DATE ISSUE BY PRINCIPAL CERTIFYING AUTHORITY, DRAWING NUMBER, DATE, REVISION NUMBER AND DESIGNER DETAILS} including all gutters, pipes, drains, walls, kerbs, pits, grates, tanks, chambers, basins and surfaces designed to temporarily detain stormwater on the land.

The Act means the Conveyancing Act 1919.

A copy of the construction certificate or complying development certificate is held on Council file "DA number" or "CDC number".

Name of authority having the power to release, vary or modify the 'Positive Covenant' is City of Canada Bay Council.

Restriction on the Use of Land

Existing Allotments

- A3 The Terms of Restriction on the Use of Land - Where there is no land subdivision (no Section 88B instrument required) then the following standard wording is to be used for the "The Terms of Restriction on the Use of Land" and attached with the standard NSW Department of Lands form 13RPA.

Terms of Restriction on the Use of Land

(Show full details of the Restriction)

The registered proprietors covenant with the City of Canada Bay Council (Council) that they will not:

- I. Do any act, matter or thing which would prevent the structure and works from operating in an efficient manner.
- II. Make any alterations or additions to the structure and works or allow any development within the meaning of the Environmental Planning and Assessment Act 1979 to encroach upon the structure and works without the express written consent of the authority.
- III. This covenant shall bind all persons who claim under the registered proprietors as stipulated in Section 88E(5) of the Act.

For the purposes of this covenant:

Structure and Works shall mean the on-site stormwater detention (OSD), on-site stormwater retention/absorption (OSA), stormwater quality improvement device (SQID) and/or mechanical pump-out system constructed on the land as set out in the plan annexed hereto and marked with the letter "A" or alternatively as detailed on the plans approved by the Principal Certifying Authority (INSERT ORGANISATION/COMPANY NAME): {INSERT CONSTRUCTION CERTIFICATE NUMBER AND DATE ISSUE BY PRINCIPAL CERTIFYING AUTHORITY: {INSERT DA NUMBER, DRAWING NUMBER, DATE, REVISION NUMBER AND DESIGNER DETAILS} including all gutters, pipes, drains, walls, kerbs, pits, grates, tanks, chambers, basins and surfaces designed to temporarily detain stormwater on the land.

The Act means the Conveyancing Act 1919.

Name of Authority having the power to release, vary or modify the 'Restriction' is City of Canada Bay Council.

Land Subdivision

- A4 The Terms of Restriction on the Use of Land - Where a subdivision has been lodged and a Section 88B instrument created, then the following standard wording for the "The Terms of Restriction on the Use of Land" shall be used.

Terms of Restriction on the Use of Land referred to in the above-mentioned Plan

The registered proprietor covenant with the City of Canada Bay Council (Council) in respect to the structure erected on the land described as "on-site stormwater detention system" (which expression includes all ancillary gutters, pipes, drains, walls, kerbs, pits, grates, tanks, chambers, basins and surfaces designed to temporarily detain stormwater) shown on plans approved by the Principal Certifying Authority: {INSERT DA NUMBER, DRAWING NUMBER, DATE, REVISION NUMBER AND DESIGNER DETAILS} (hereinafter called "the system").

The registered proprietors covenant with the City of Canada Bay Council (Council) that they will not:

- I. Do any act, matter or thing which would prevent the structure and works from operating in an efficient manner.
- II. Make any alterations or additions to the structure and works or allow any development within the meaning of the Environmental Planning and Assessment Act 1979 to encroach upon the structure and works without the express written consent of the authority.
- III. This covenant shall bind all persons who claim under the registered proprietors as stipulated in Section 88E(5) of the Act.

For the purposes of this covenant:

Structure and Works shall mean the on-site stormwater detention (OSD), on-site stormwater retention/absorption (OSA), stormwater quality improvement device (SQID) and/or mechanical pump-out system constructed on the land as set out in the plan annexed hereto and marked with the letter "A" or alternatively as detailed on the plans approved by the Principal Certifying Authority (INSERT ORGANISATION/COMPANY NAME): {INSERT CONSTRUCTION CERTIFICATE NUMBER AND DATE ISSUE BY PRINCIPAL CERTIFYING AUTHORITY}, including all gutters, pipes, drains, walls, kerbs, pits, grates, tanks, chambers, basins and surfaces designed to temporarily detain stormwater on the land.

The Act shall mean the Conveyancing Act 1919.

Name of Authority having the power to release, vary or modify the 'Restriction' is City of Canada Bay Council.

Rhodes Peninsula Site Specific Requirements

Minimum Standards

- MS1 General - The minimum engineering standards given in this Appendix applies to all proposed developments in the Rhodes Peninsular area.
- The general standards given in the main body of this Engineering Development Control Plan will still apply where no reference or detail is given in this Appendix.

Shared Paths

- SP1 Shared cycle ways and footpaths - shared cycle ways and footpaths shall be provided in all public domain areas. They shall be constructed to the following standards:
- Wearing surface material shall be of concrete 100mm thick, 32MPa compressive strength and reinforced with F72 mesh placed centrally.
 - Concrete pavement shall be of uniform colour to match existing area. Coloured black oxide may be used.
 - Concrete pavement to have keyed or dowelled joints at 12 metre centres. Control joints shall be saw cut at 3 metre intervals as soon as the concrete has set sufficiently as to not pull stones when cutting.
 - Concrete pavement shall be laid on a 20mm sand blinding layer on a compacted sub-grade in accordance with the **AUSPEC** specification.
 - Concrete pavement shall be finished with a coving trowel.

Seawalls

- SE1 Serviceability Level - Seawalls shall be constructed to a serviceable level in accordance with the Australian Standards for Marine Structures.
- SE2 Sea Level Rise - Seawalls shall be designed and constructed to withstand a 1%AEP storm event with no overtopping inclusive of additional height of 900mm minimum to account for sea level rise due to climate change.

Jetty

- JE1 Prohibited - Jetties or similar structures are prohibited.
- Council is unable to maintain and renew jetties due to financial constraints.

Public Domain Lighting

- ST1 Design Requirements – All public domain lighting shall be designed and constructed to the requirements and specification of the City of Canada Bay Council.

Tree Planting

- TP1 General - Refers to tree planting in road carriageways and footways in general.
- TP2 Street Trees - Street trees generally shall be planted within a square reinforced concrete cut-off wall as a structural root barrier. Approved synthetic root barrier systems or equivalent may be used in footway areas (pedestrian access areas only).
- Where street trees are to be planted in road carriageways (pavement area subject to vehicular loads), it shall be planted within a square reinforced concrete cut-off wall only.
- The root barriers shall be extended 300mm below the road pavement. The purpose of the root barriers is to protect the road pavement from moisture and root ingress.

Design Life of all Structures

- DL1 General - All public infrastructure shall be designed and selected to provide the least lifecycle costs with respect to maintenance and renewal of the infrastructure throughout its entire serviceable life.
- DL2 Seawalls - Seawalls shall be designed and constructed for a minimum design life of 100 years in accordance with the Australian Standards for Marine Structures.

Civic Signage

- SG1 General - All signage shall be designed and installed in accordance with Canada Bay Council's Signage Manual.

Garbage Bins

- GR1 General - All garbage bins shall be minimum 240 Litres wheelie bin type and housed in stainless steel and timber surround (model no. EM235 Bennelong Bin Enclosures by Emerdyn Pty Ltd).

Stormwater Drainage

- SD1 General - Stormwater drainage systems shall be designed generally in accordance with Section **ES3 Stormwater Management**.
- SD2 On-site Stormwater Detention - On-site stormwater detention (OSD) shall be required for all proposed development allotments unless drainage from the developed site is discharged directly into the Bay, or that the street trunk drainage system has been designed to cater for the 1%AEP storm event.
- Where OSD is required, the permissible site discharge shall be based on a "greenfields" site during 20%AEP with an impervious area of 0%.
- SD3 Sub-surface Water - Sub-surface water collected from basements and all lowered floor areas shall not be discharged directly to the kerb. Refer to Section **ES3 Stormwater Management, Sub-surface Water** for further guidelines.
- SD4 Scour and Erosion Control - Refer to Section **ES3 Stormwater Management** for further guidelines.

Works as Executed Drawings and Asset Data

- WAE1 Works-As-Executed Drawings - Prior to handing over land or assets to Council, the developer is to provide Works-As-Executed drawings prepared and certified by a registered surveyor.

The plans are to be provided in the form of:

- A1 size pdf drawings suitable for archiving (vector based, not raster) and
- A vector format suitable for importing into Council's Land Information Systems (MapInfo) or Cad systems (AutoCAD).

- WAE2 Detailed Asset Data - Prior to handing over land or assets to Council, the developer is to provide detailed asset data at a component level for importation into Council's Asset Management System.

Information required will depend on the components and the developer should submit a component list to Council for further advice.

The following list is representative of what would be required:

- A vector representation of the component and its location on MGA94 coordinates linked to:
 - Description.
 - Material details.
 - Design life.
 - Installation date.
 - Acquisition date.
 - Dimensional and quantity information where it is not defined by the graphic object (e.g. The length of a pipe can be represented by the length of a line object on the plan. A data value representing the diameter needs to be attached to that line object. The area of a segment of road can be defined by a polygon. A data value representing the pavement depth needs to be attached to the polygon.)
 - Valuation at installation date.

It is recommended that these requirements be considered at the design documentation stage.

For more complex and proprietary items, for example pumps or lighting systems, the developer shall provide to Council's satisfaction details of Brand, Model, Supplier, Warranty, Installation Guide, User Guide, Safety Instructions, Servicing Record, Maintenance Manual, Keys etc as appropriate.

Community Land

- | | |
|--|---|
| <p>CL1 Plan of Management - This Section provides a guideline for the preparation of a Plan of Management for open space land which will be transferred to Council as community land.</p> <p>CL2 The Local Government Act - The <i>Local Government Act 1993</i> emphasises that Plans of Management should be prepared for all community land for which Council will be responsible to ensure land it owns or controls is actively and effectively managed.</p> <p>CL3 Description of Use - The Plan of Management should describe how public land will be managed, maintained and utilised, who is responsible for its management, its facilities and the uses and activities that occur there.</p> <p>CL4 Status - The plan should show the status of each park or reserve and the planning of each, meet the other requirements of the Act pertaining to community land management, have informed the community based on a consultative process and establish a framework for meeting community needs regarding open space.</p> | <p>CL5 Strategic Document - The plan would be a strategic document with prioritised actions which give Council flexibility to develop strategies consistent with the objectives identified in its Management Plan.</p> <p>CL6 Development of Opportunities - The plan should also enable Council to take advantage of any opportunities that arise during the life of the plan that enable implementation of the strategies in ways that may be more cost-effective, more time efficient, or in some other way provide benefits that were unavailable or not recognised during preparation of the plan.</p> |
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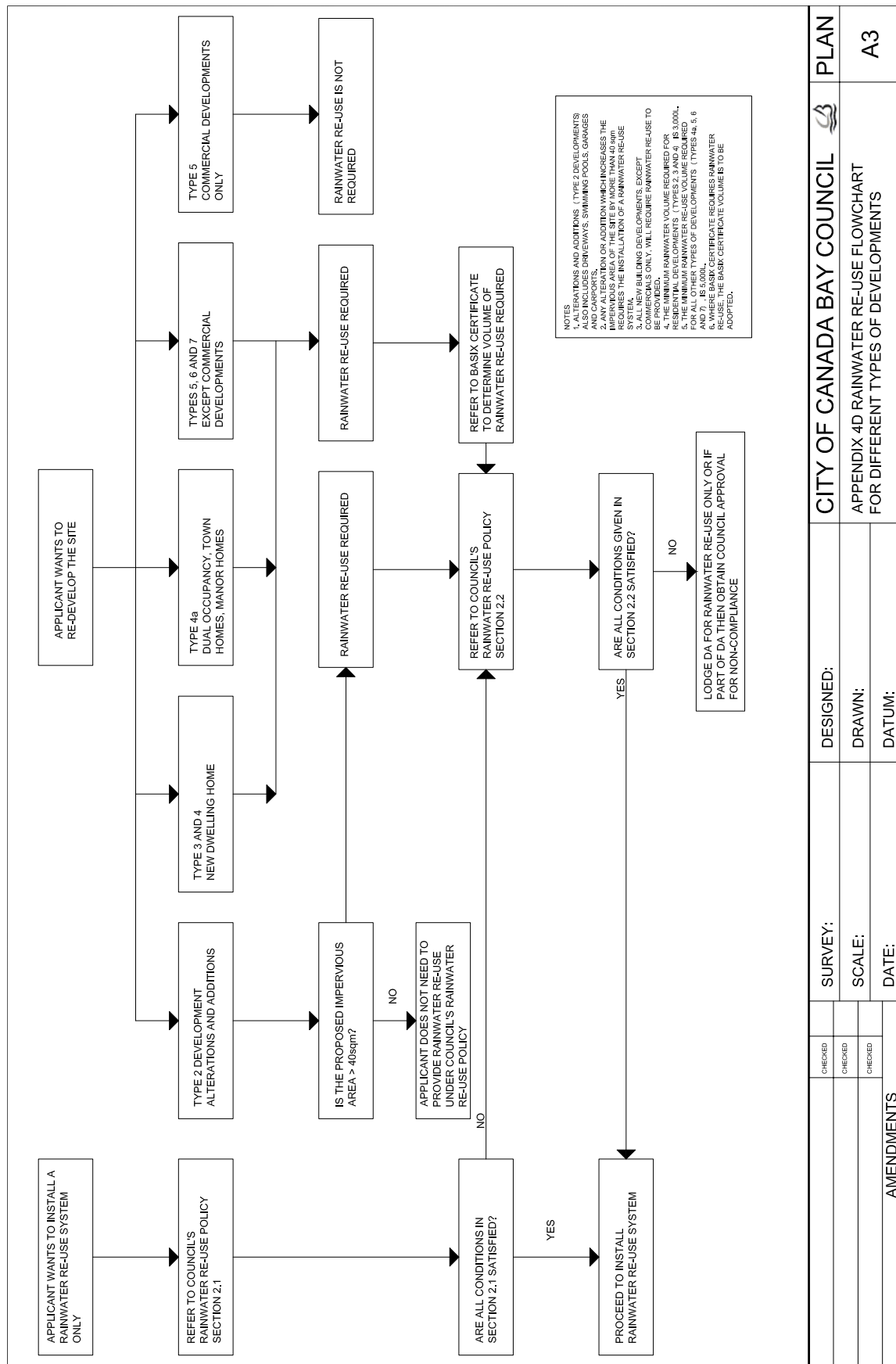
Strathfield Triangle Site Specific Requirements

Minimum Standards

- MS1 General - The minimum engineering standards given in this Appendix applies to all proposed developments in the North Strathfield area otherwise known as the Strathfield Triangle.

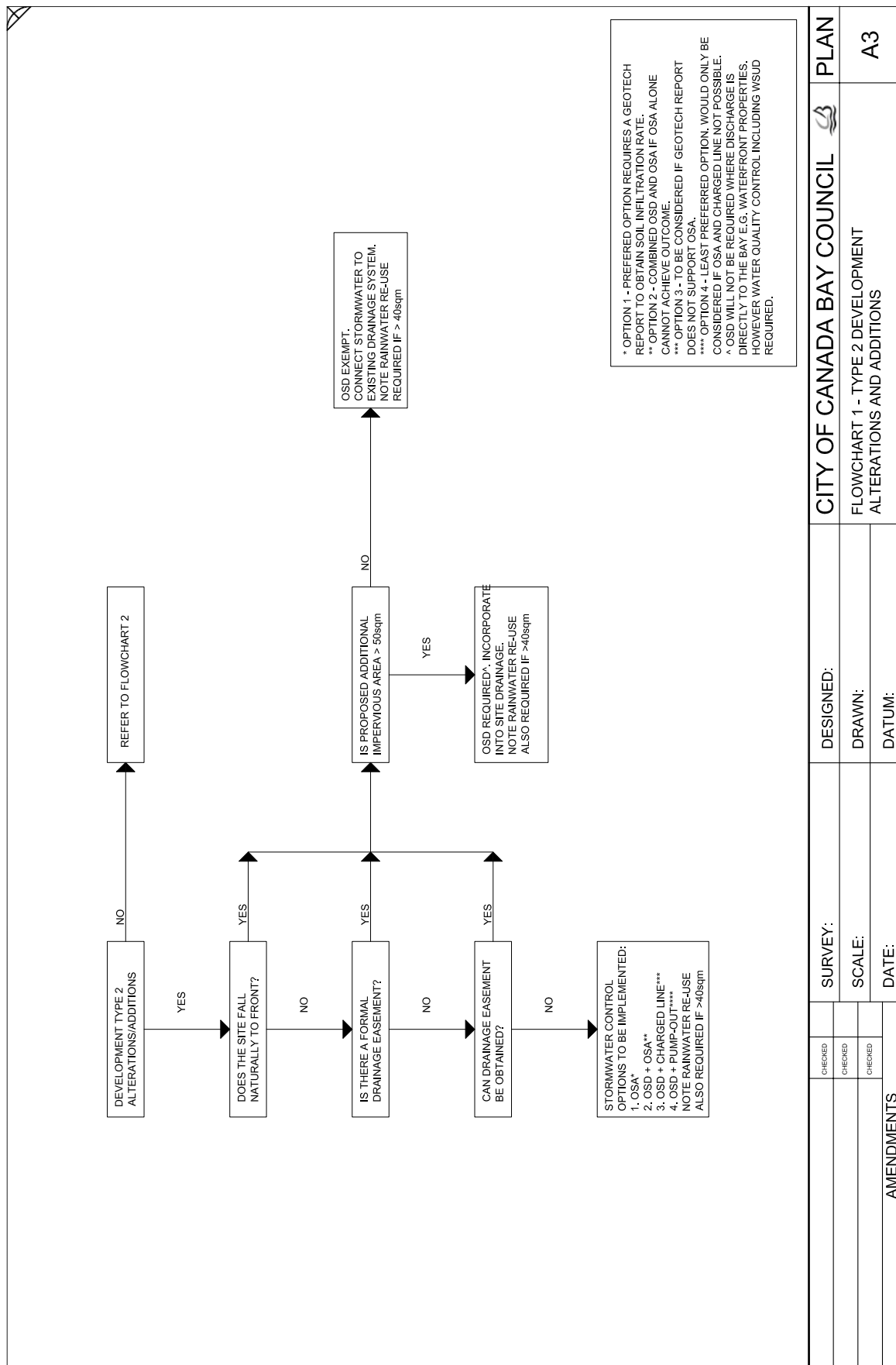
The general standards given in the main body of this Engineering Specification will still apply where no reference or detail is given in this Appendix.

Flow Charts Rainwater Reuse



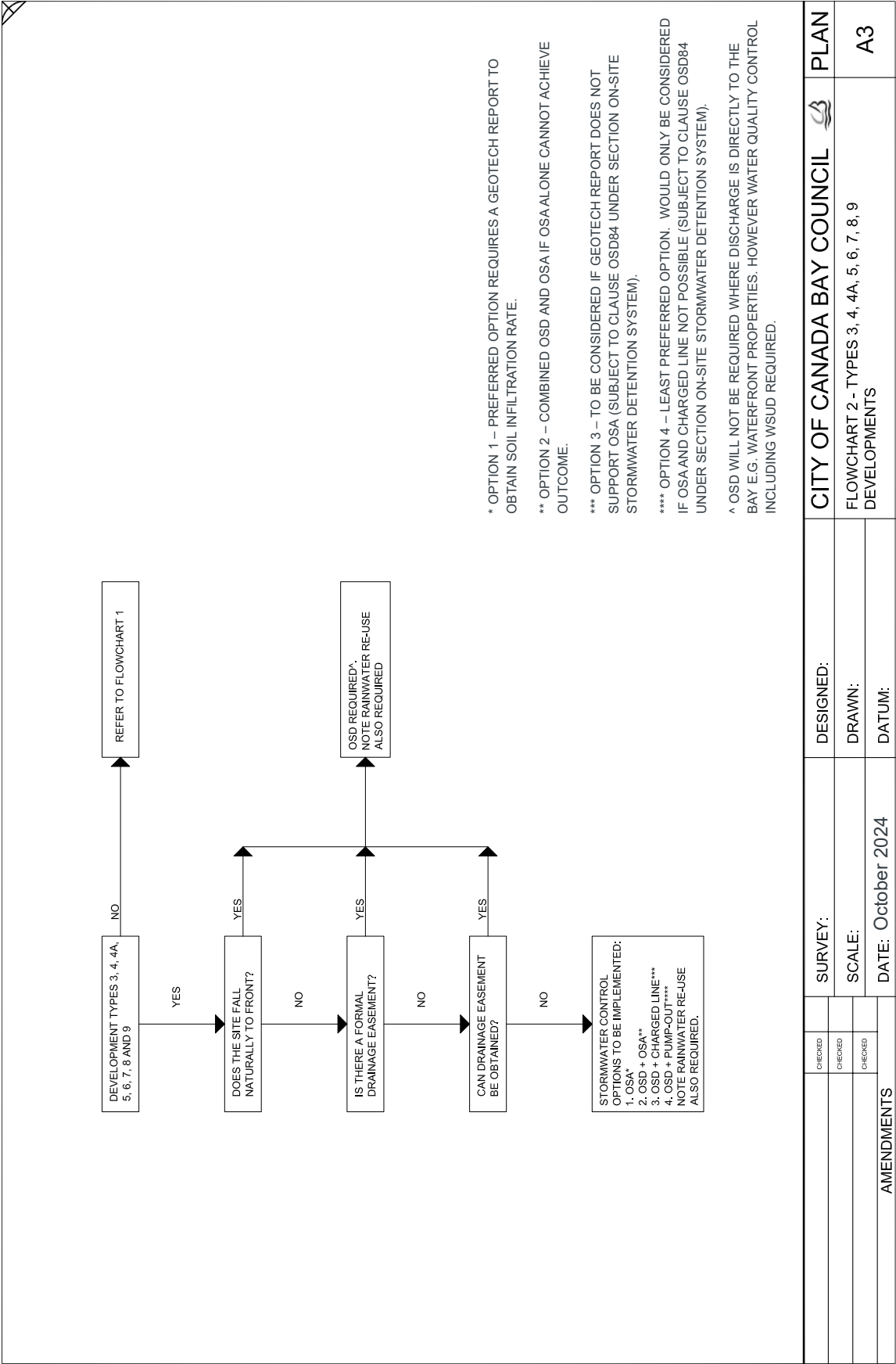
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AMENDMENTS						

Stormwater Management Type 2 Developments

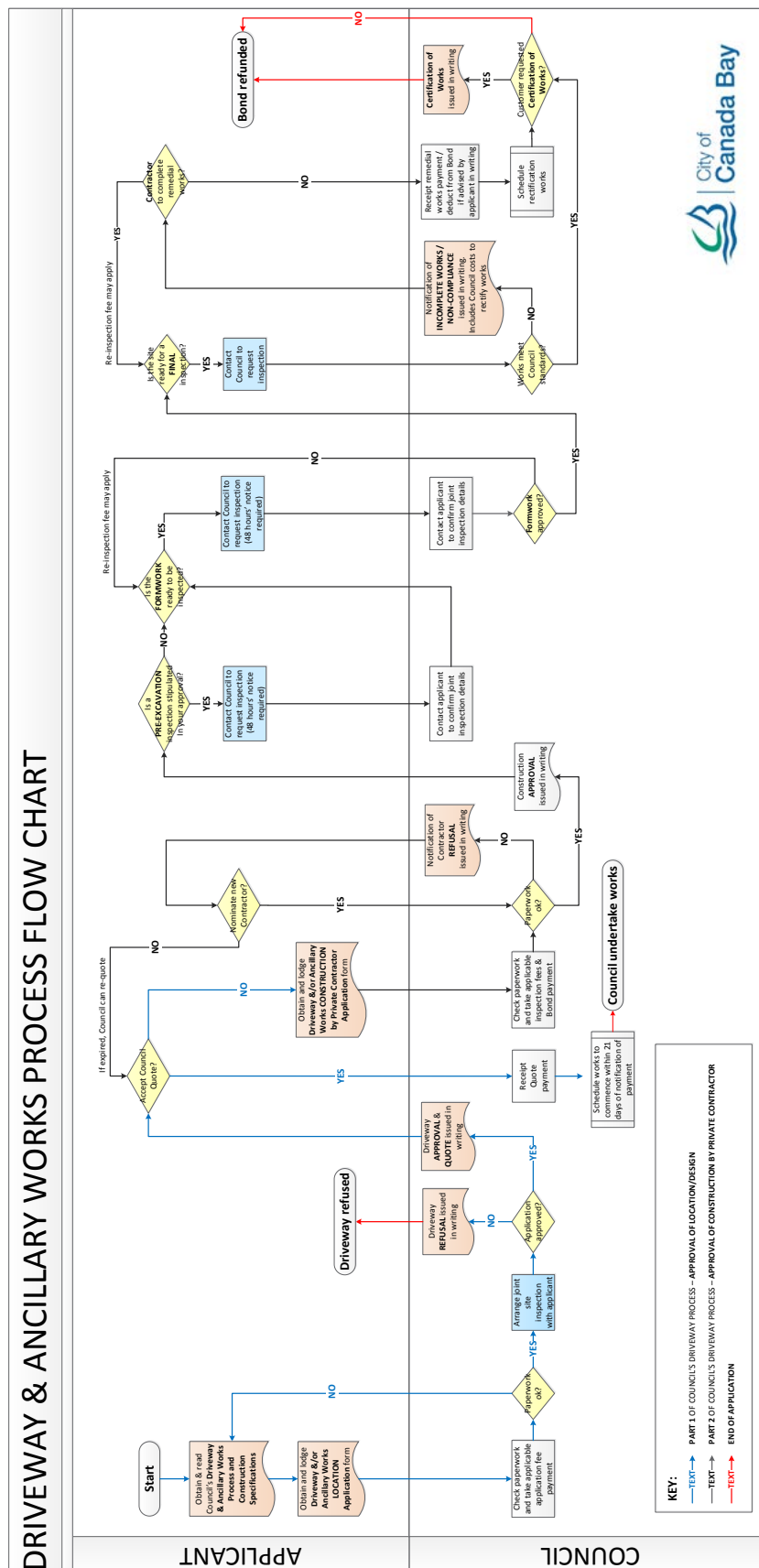


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Stormwater Management Type 3 to 9 Developments

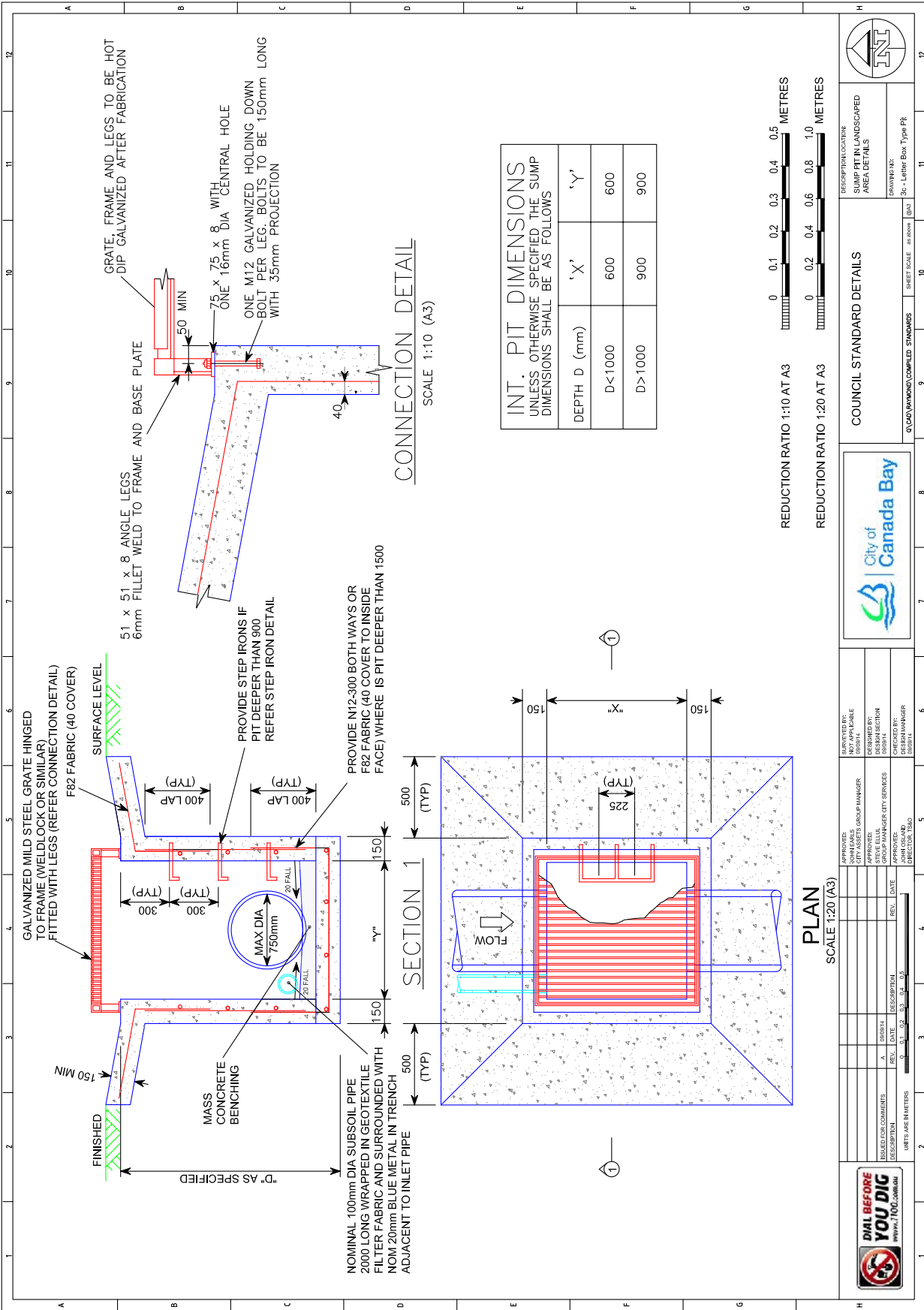


Driveway and Ancillary Works

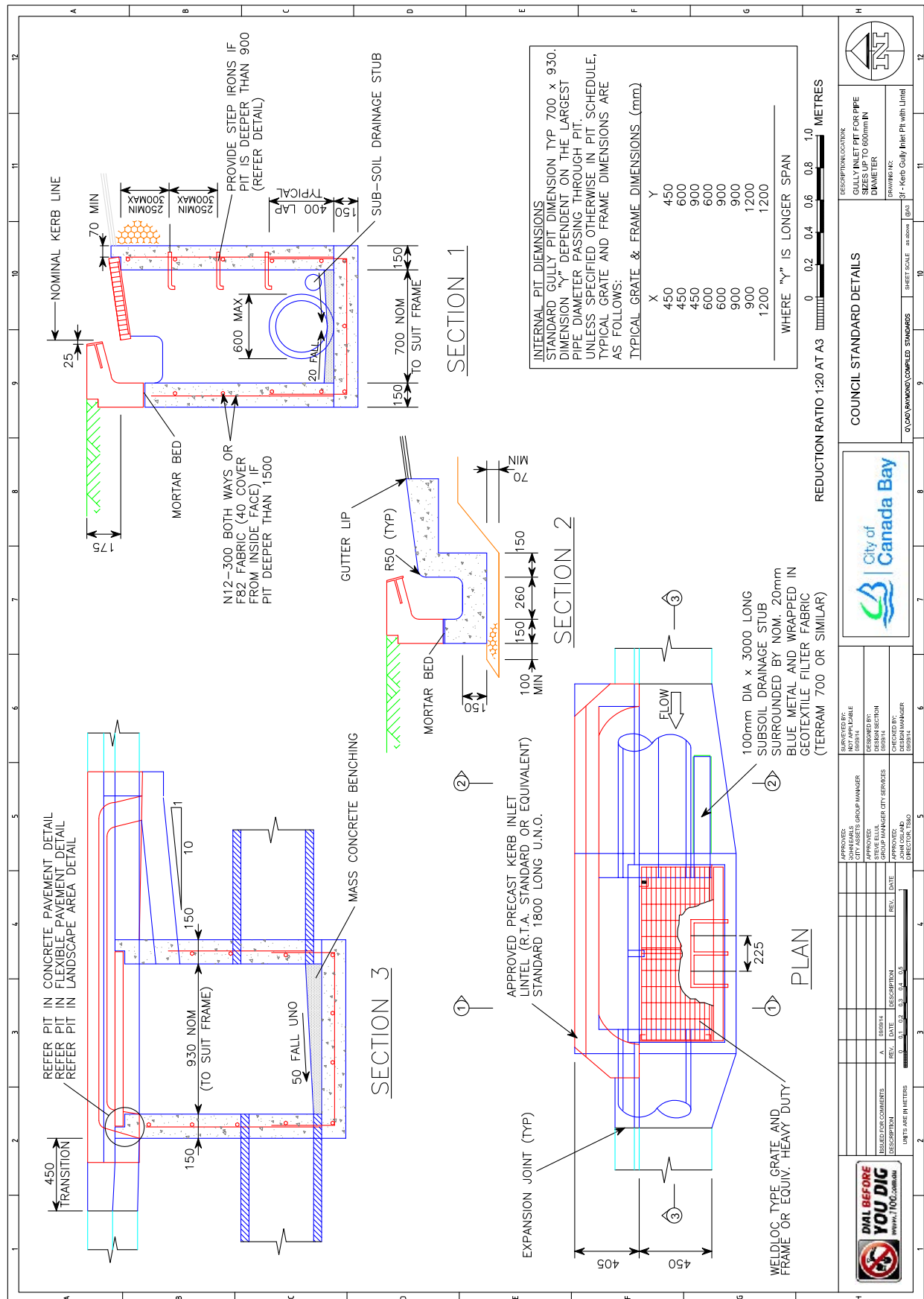




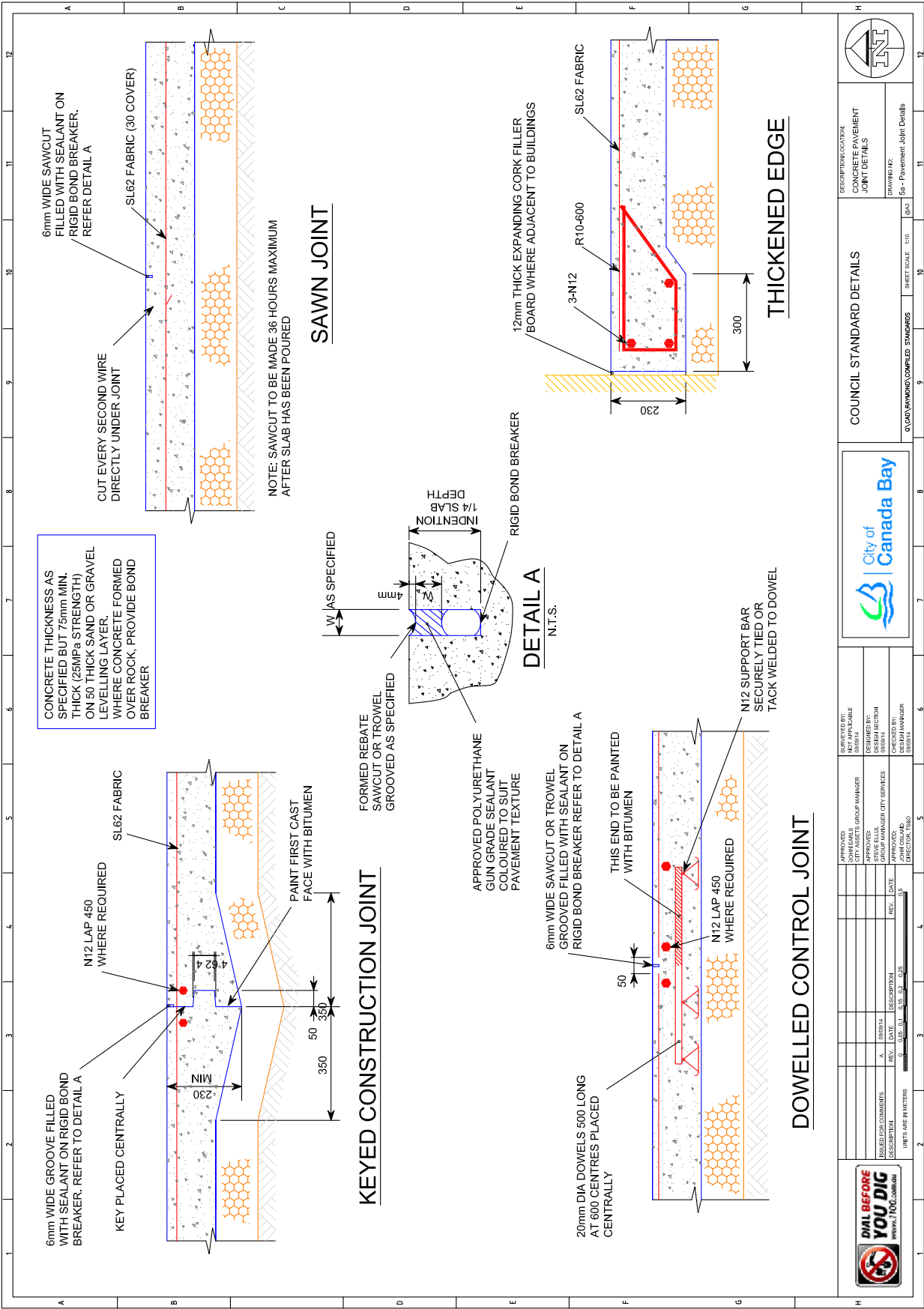


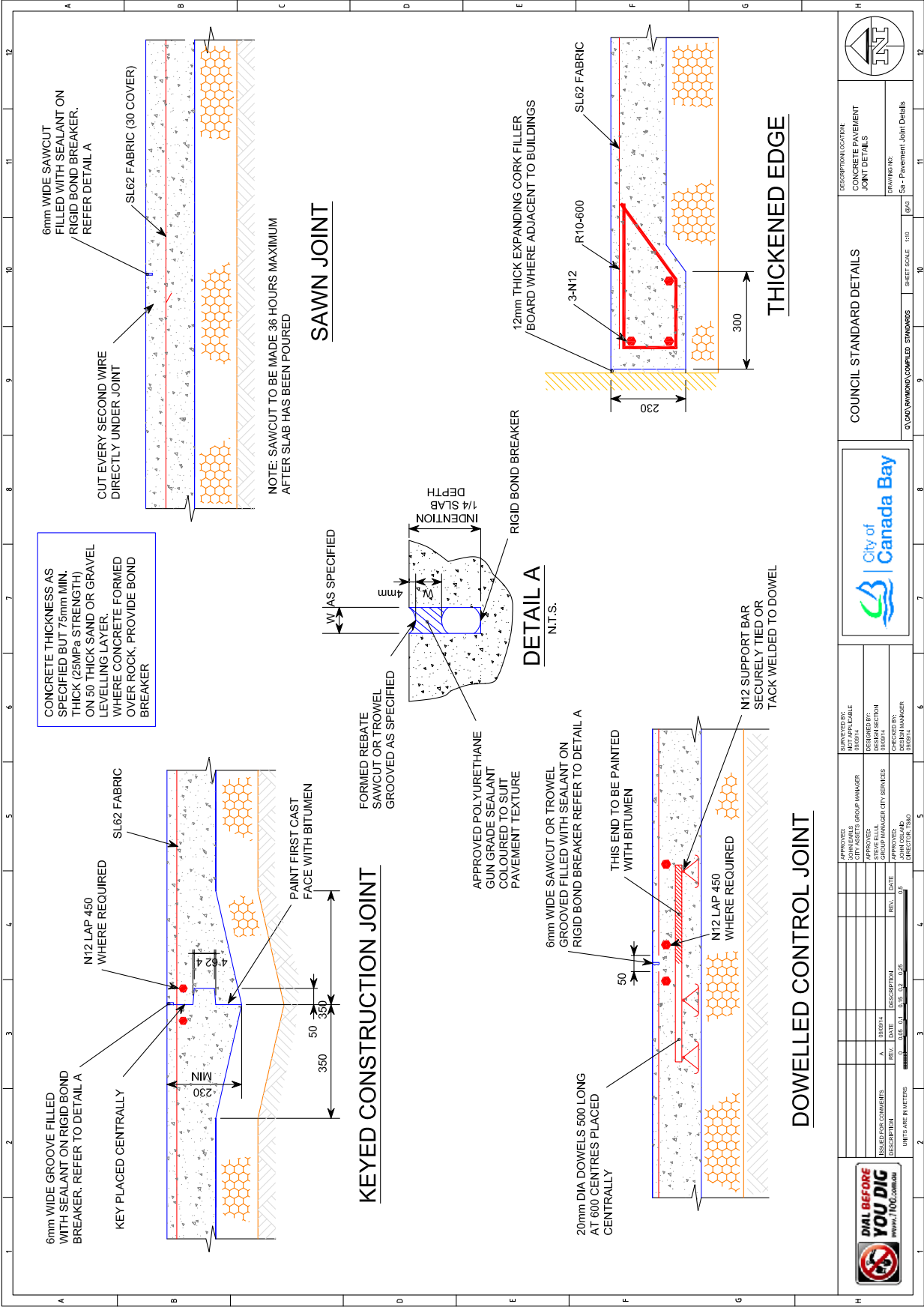












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